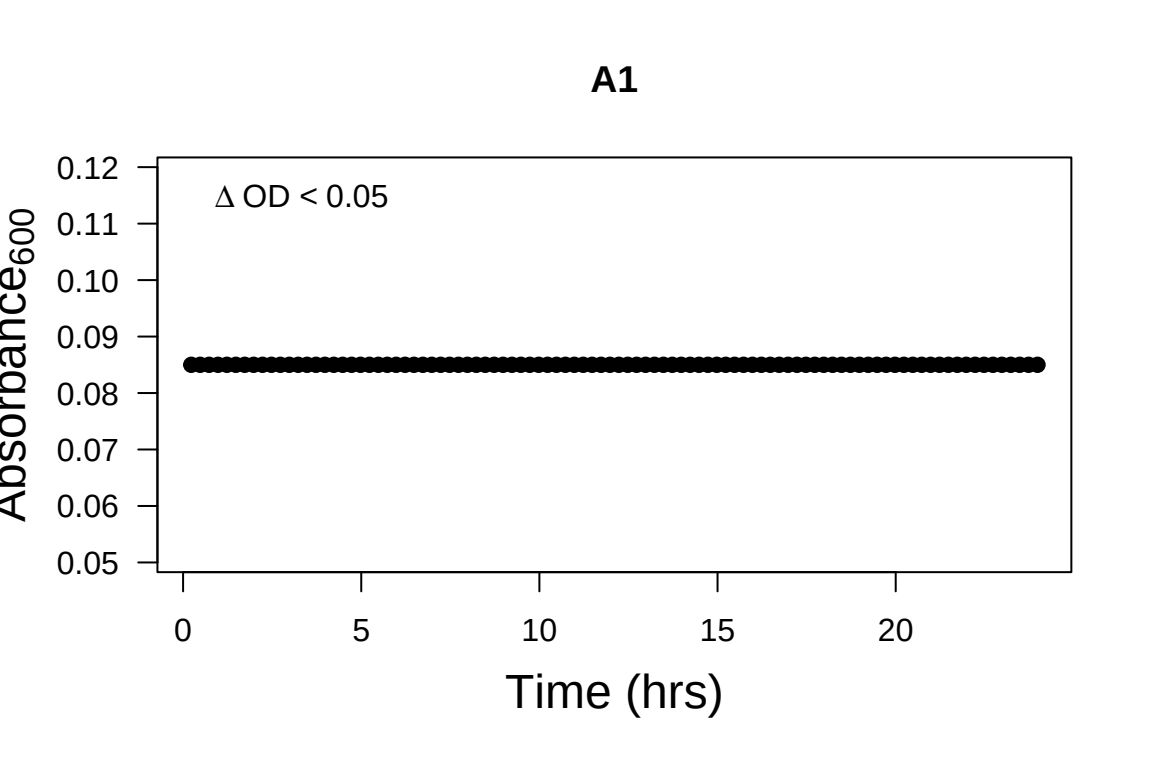
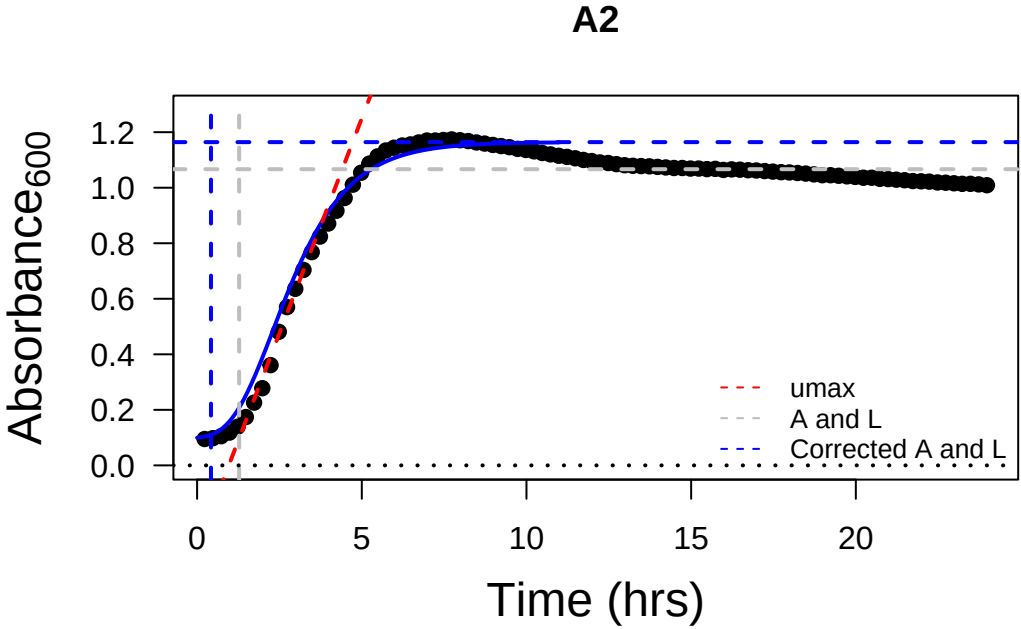
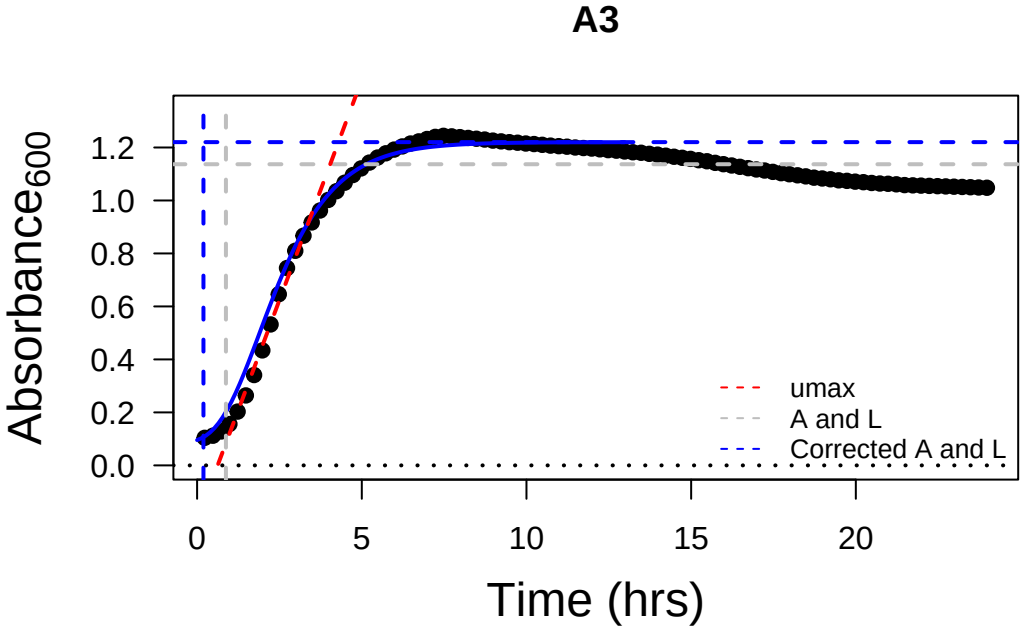
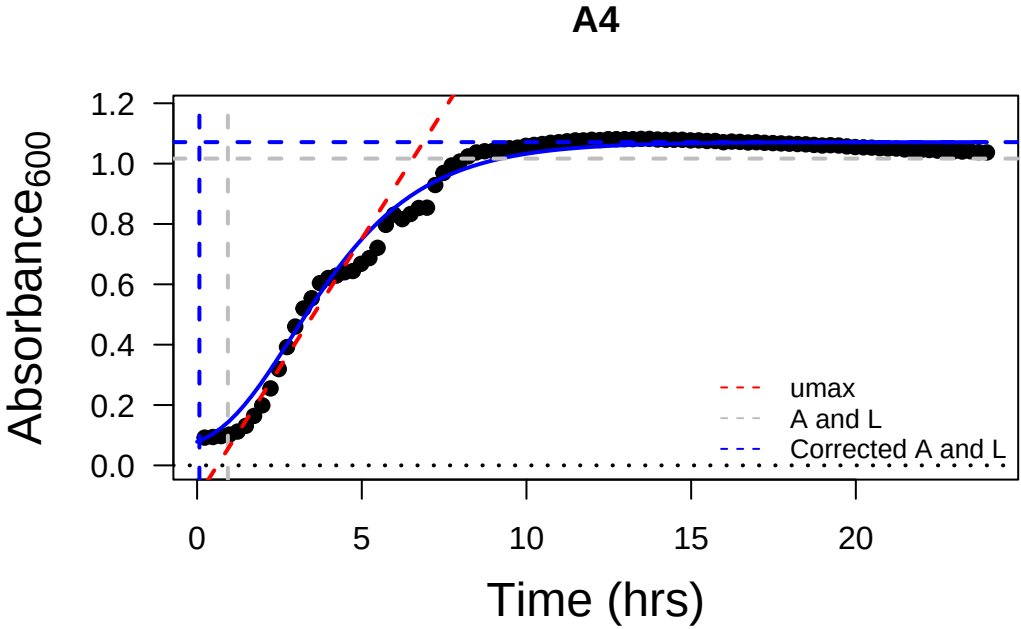
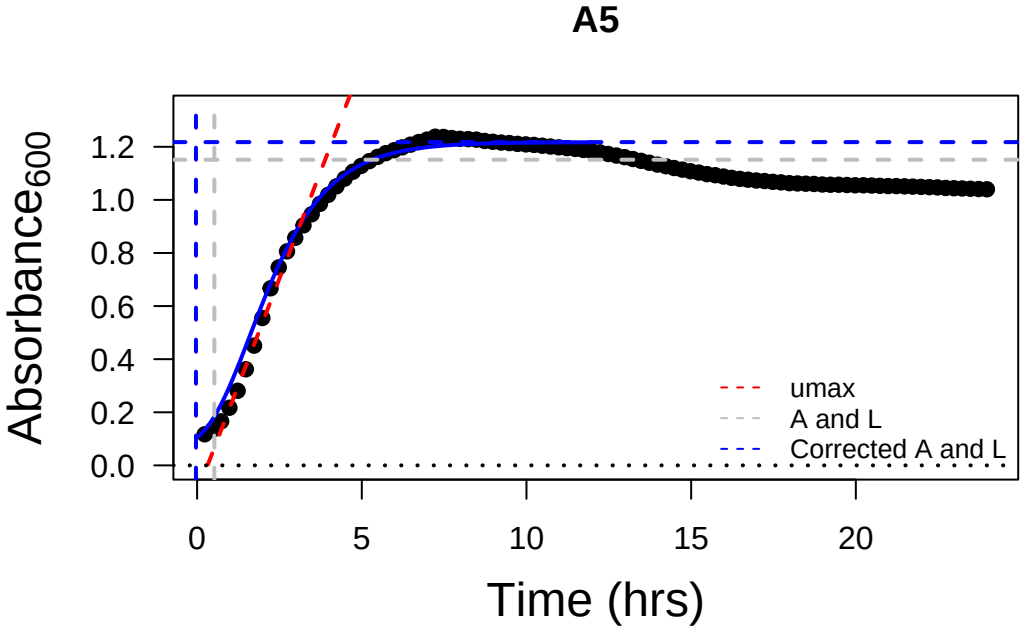
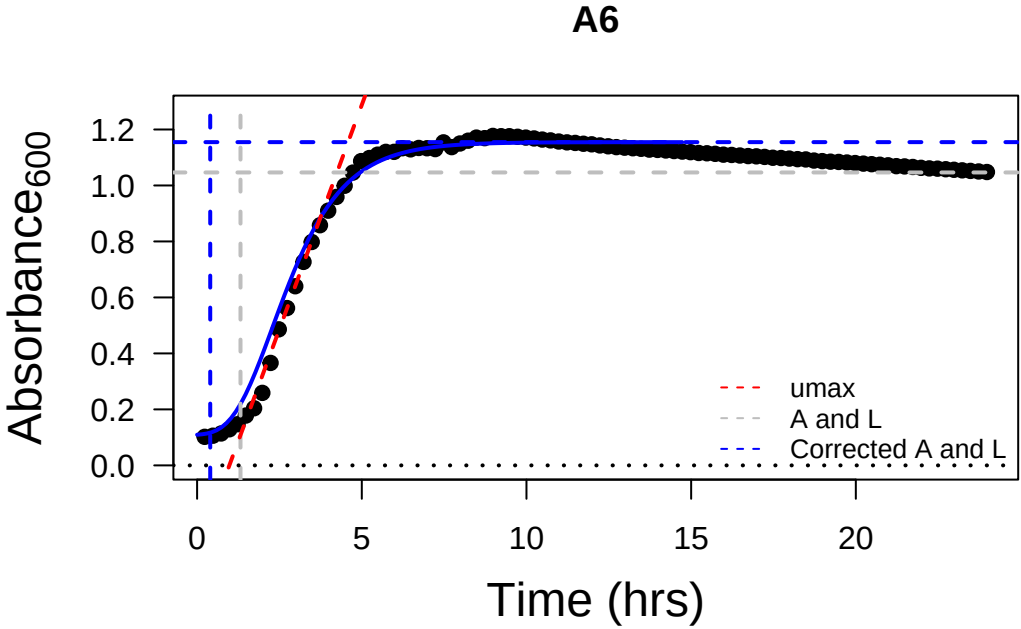

A1
Δ OD < 0.05
Absorbance600
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0
5
10
15
20
Time (hrs)

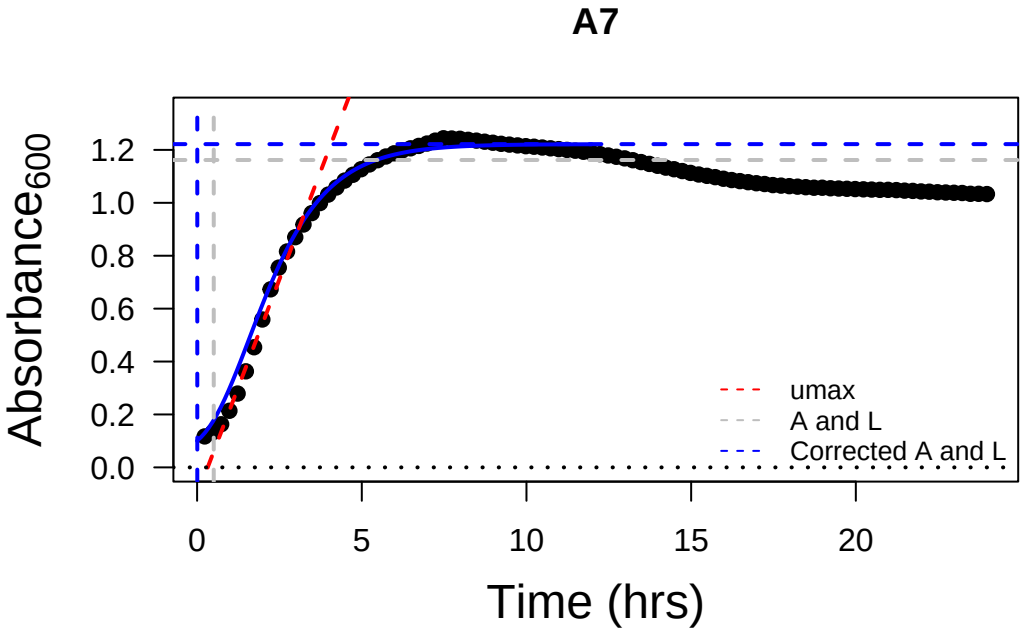

A2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

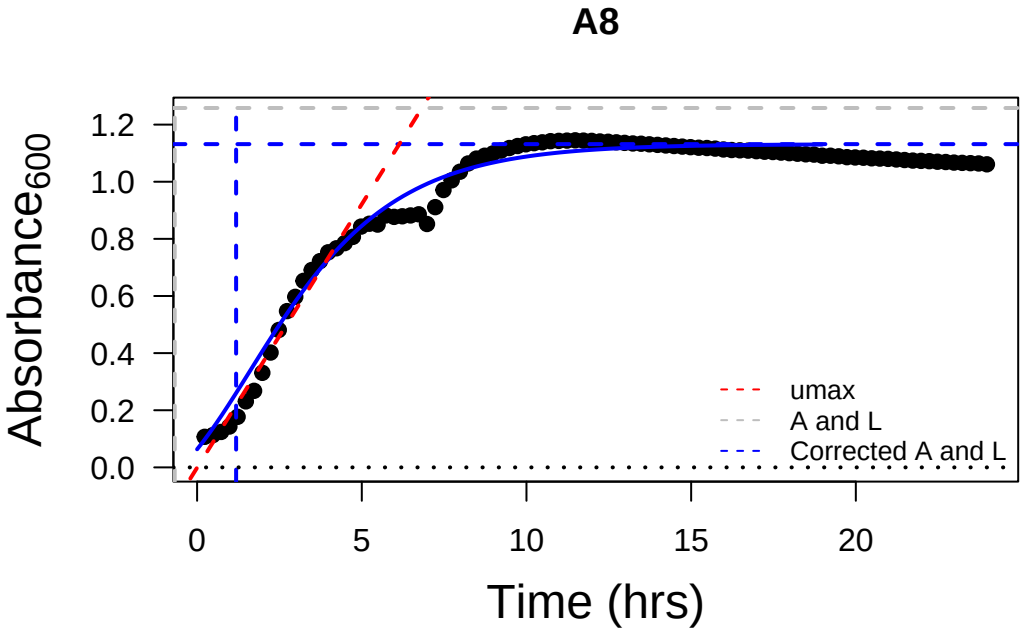

A3
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

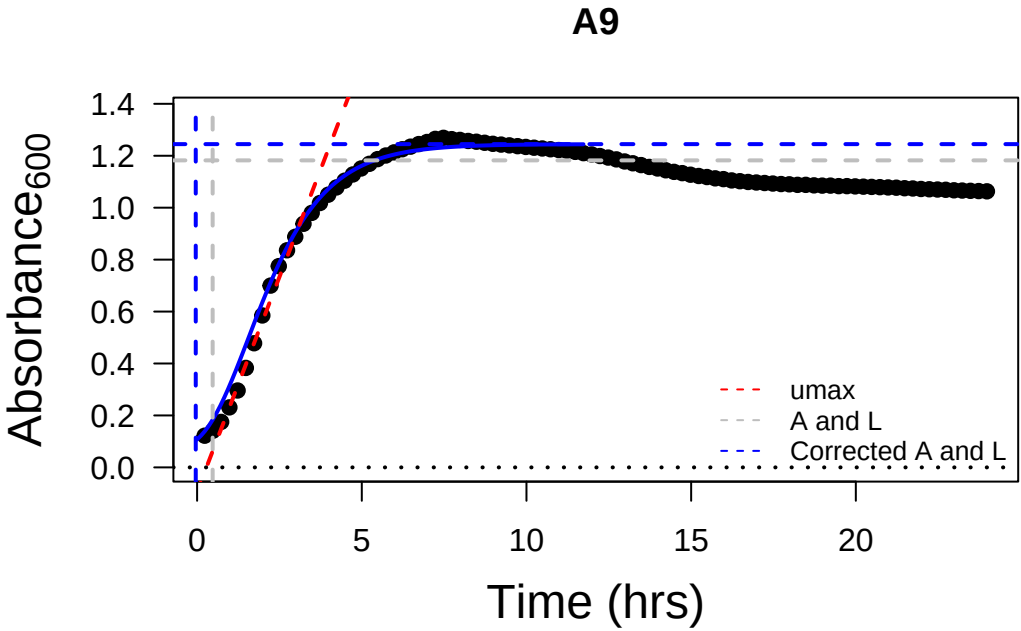

A4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

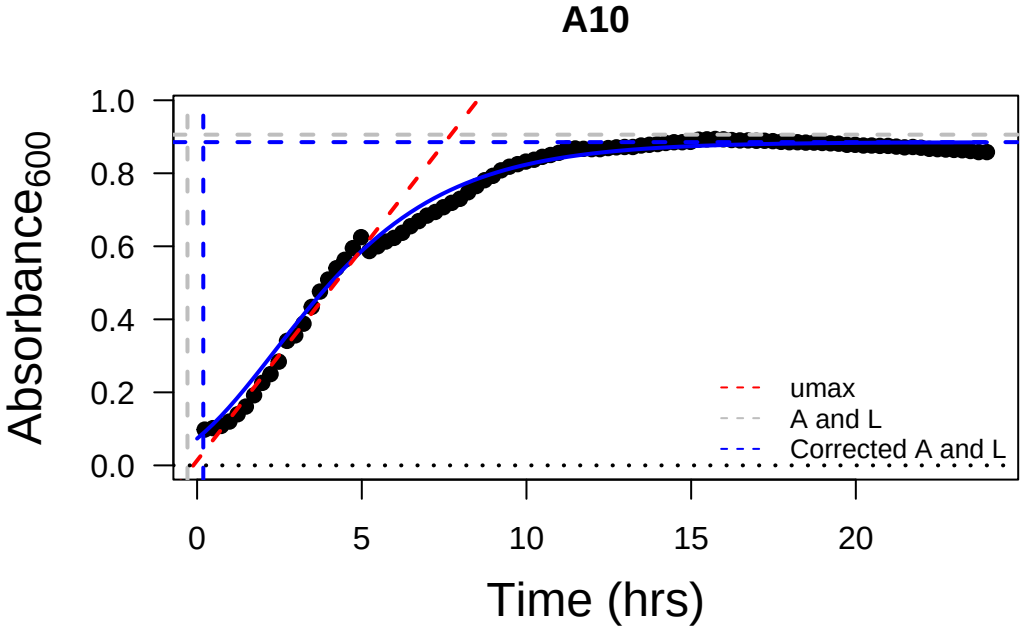

A5
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

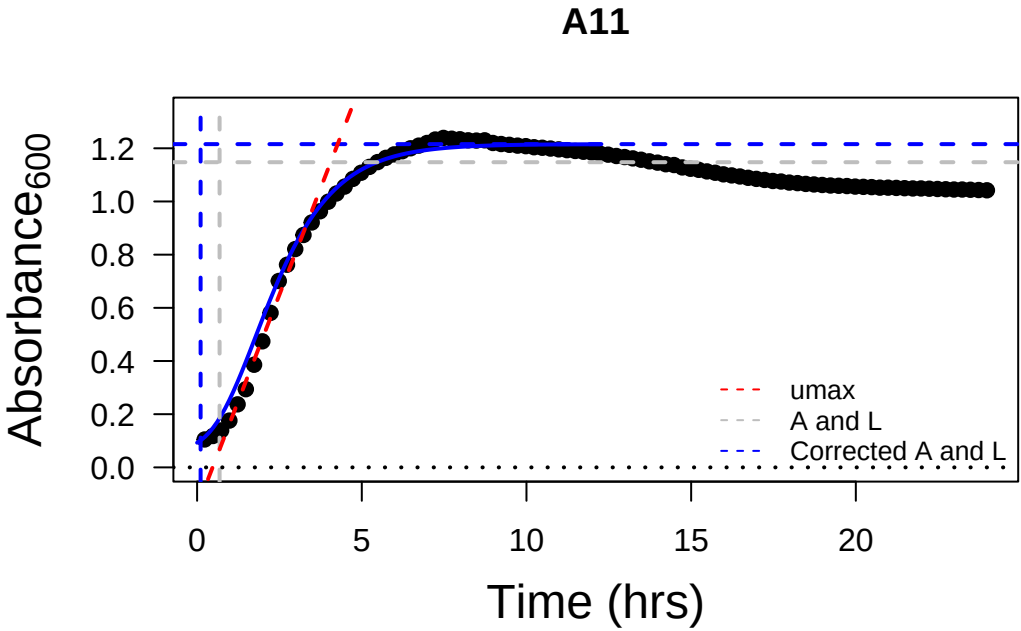

A6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

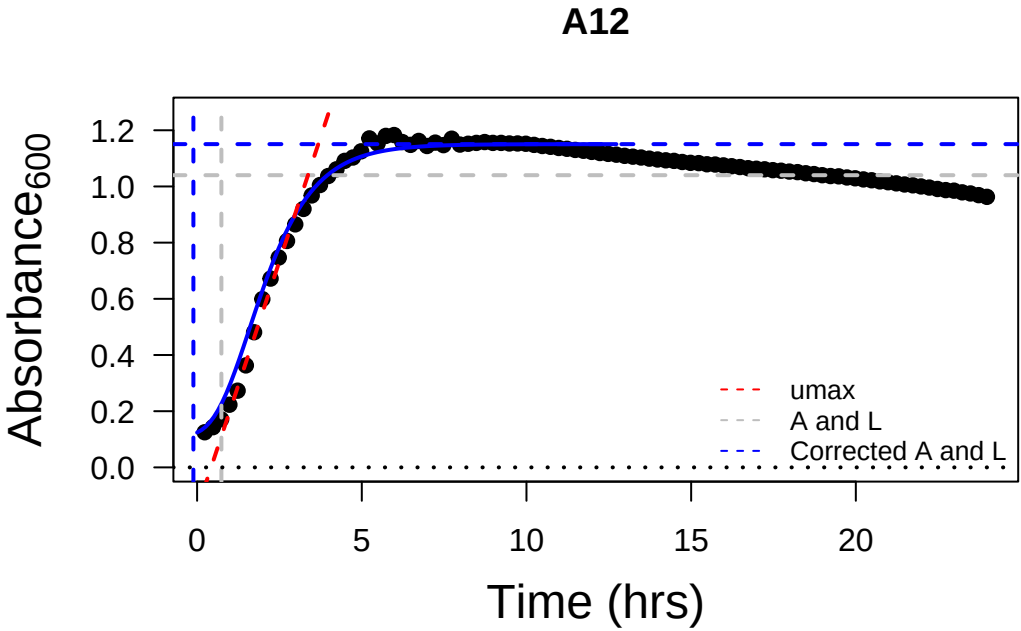

A7
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

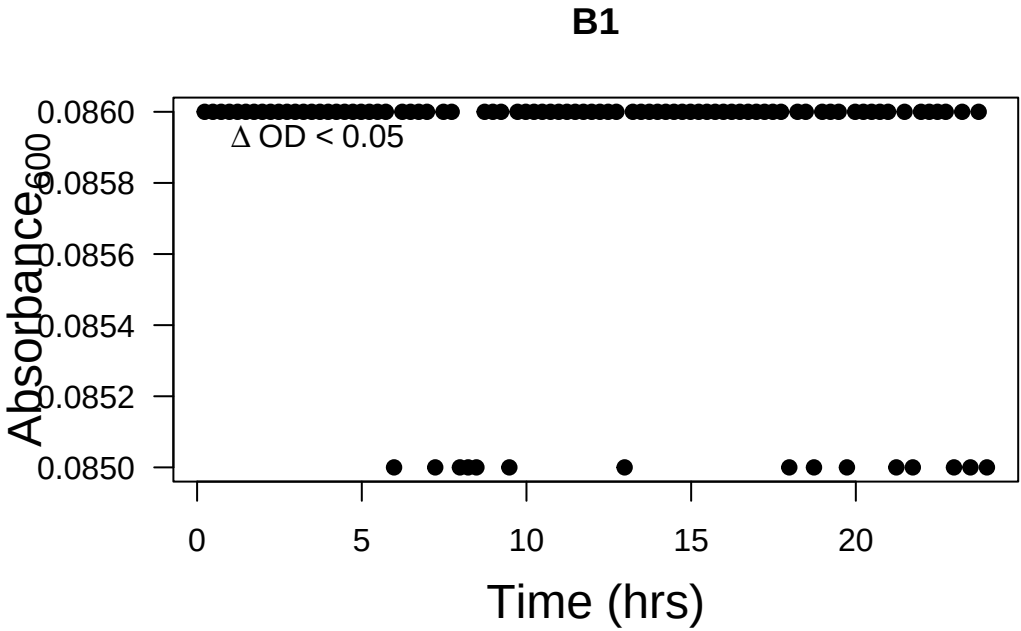

A8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

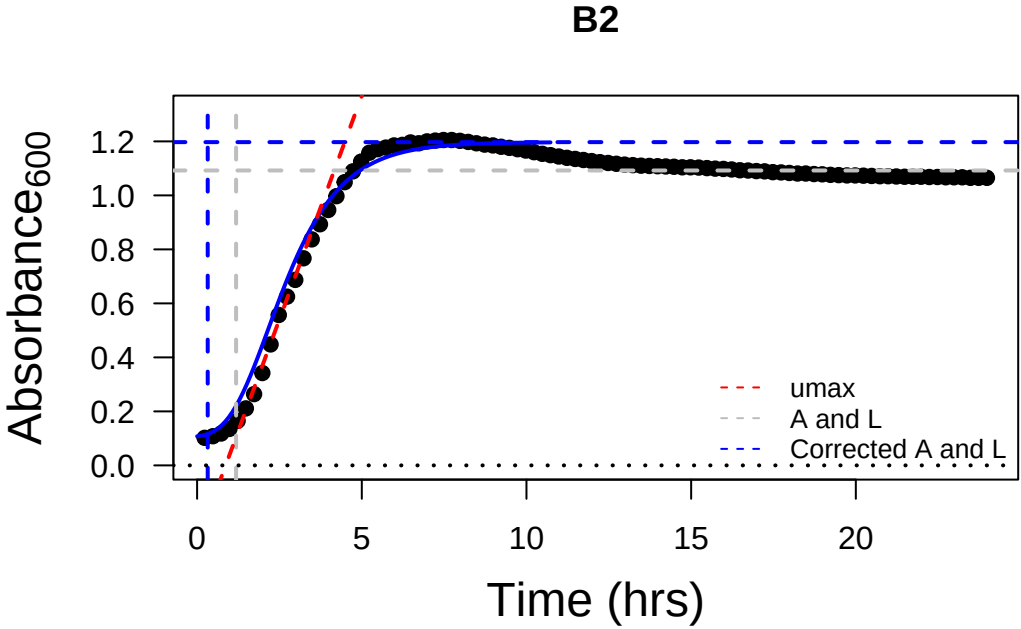

A9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

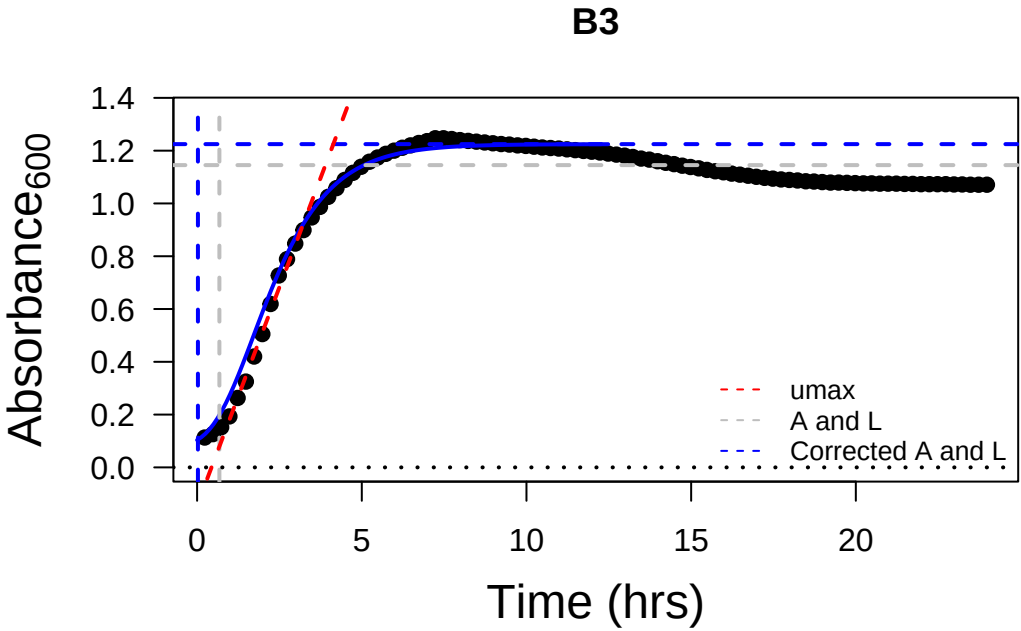

A10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

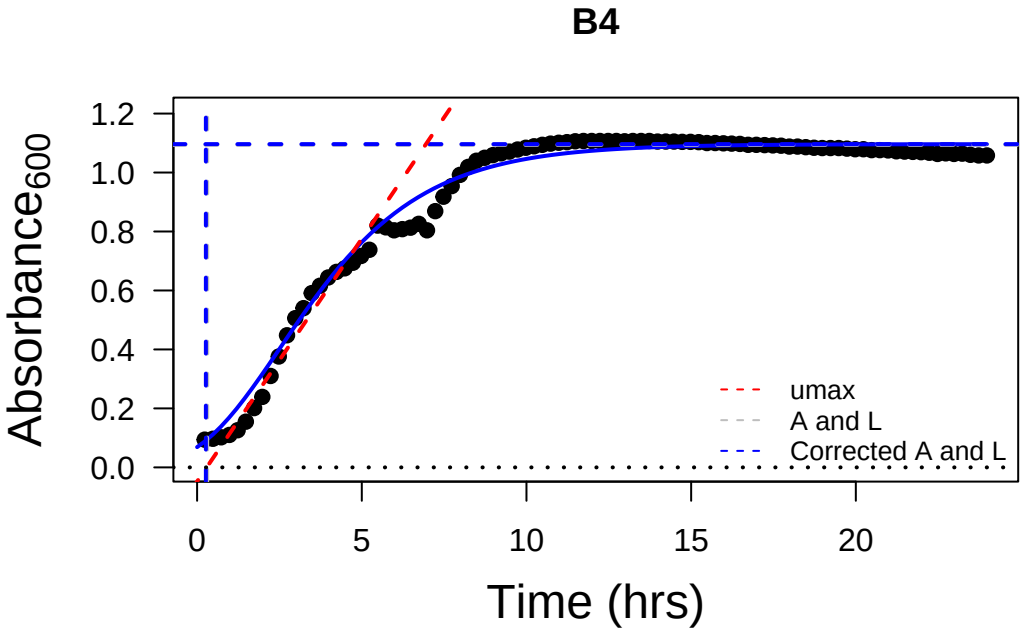

A11
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

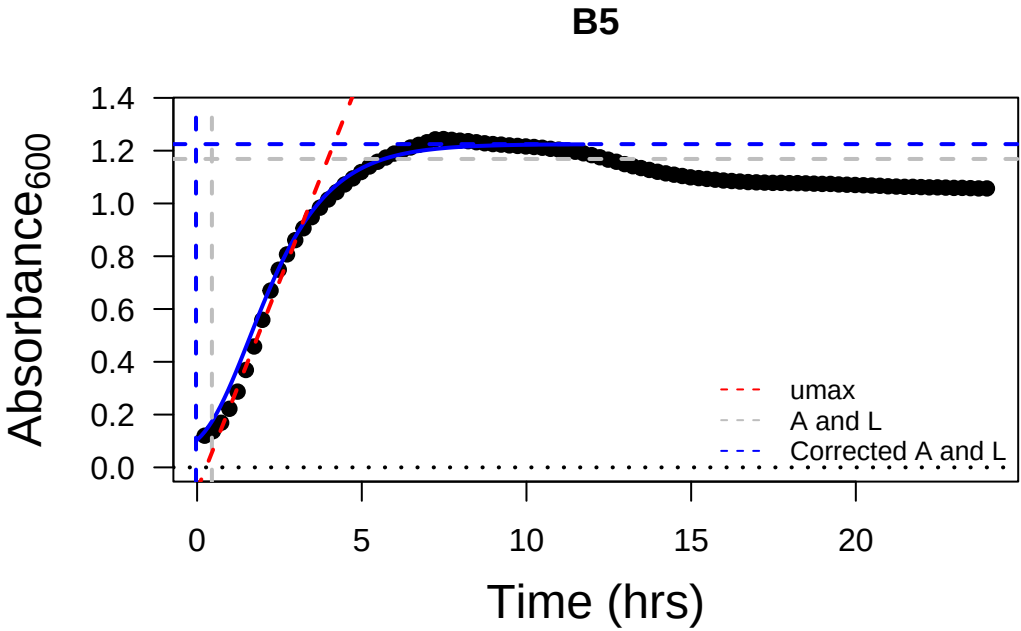

A12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

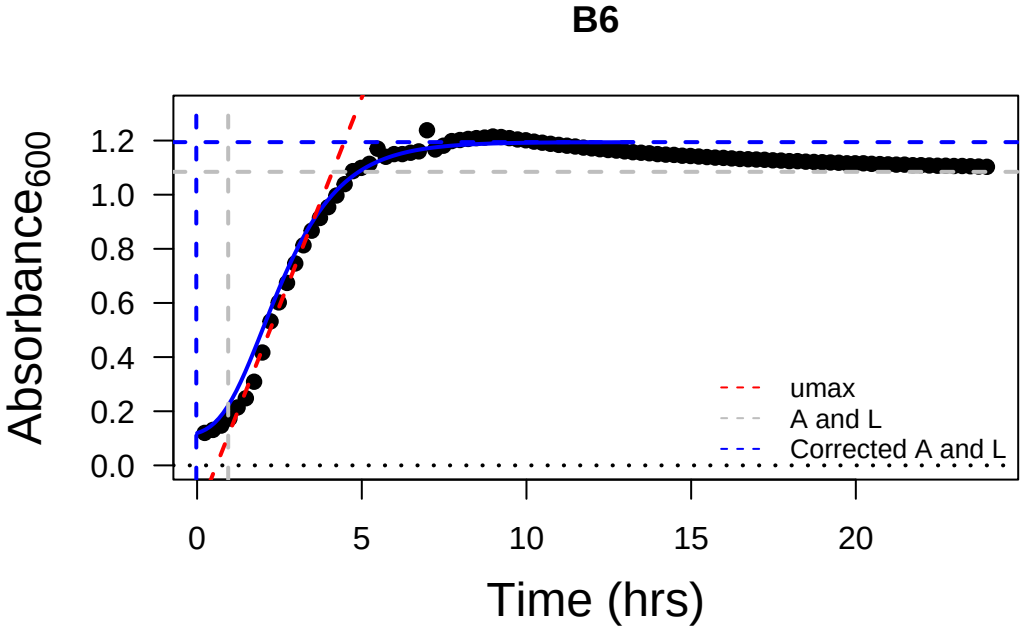

B1
Absorbance600
0.0860
0.0858
0.0856
0.0854
0.0852
0.0850
Δ OD < 0.05
0
5
10
15
20
Time (hrs)

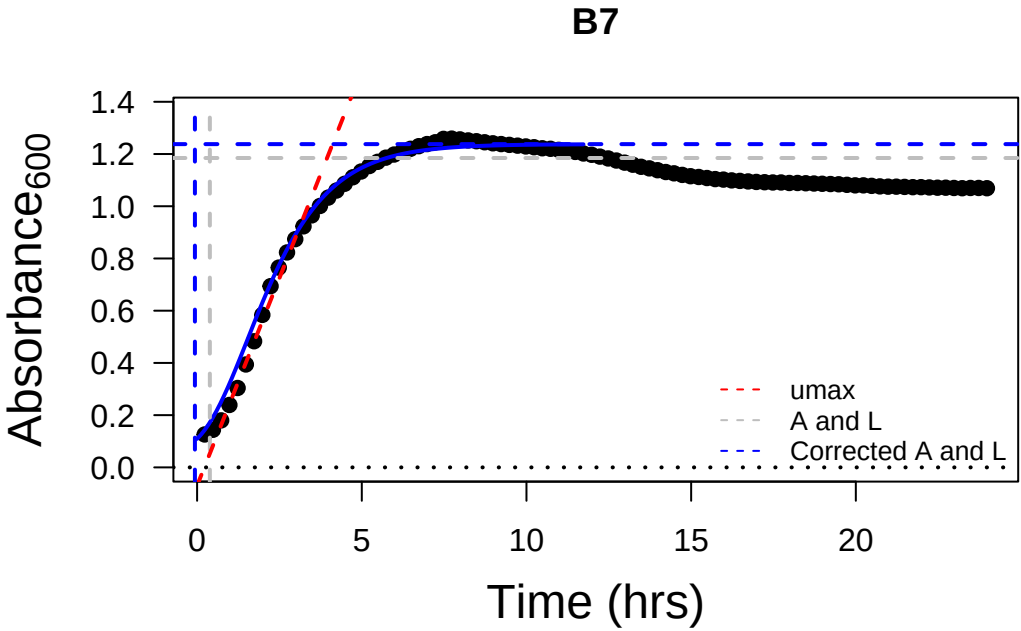

B2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

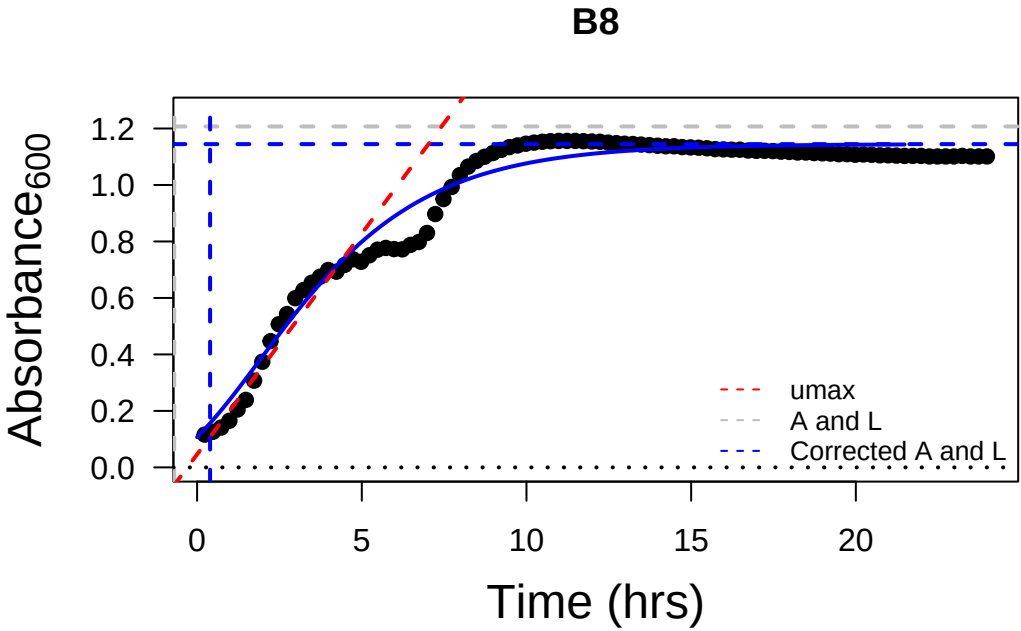

B3
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

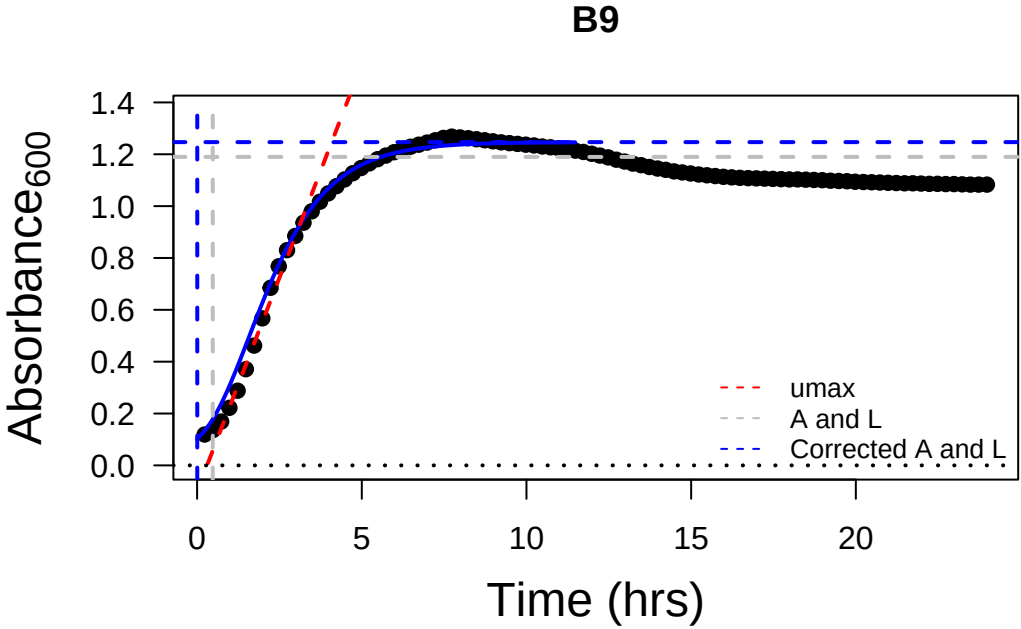

B4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

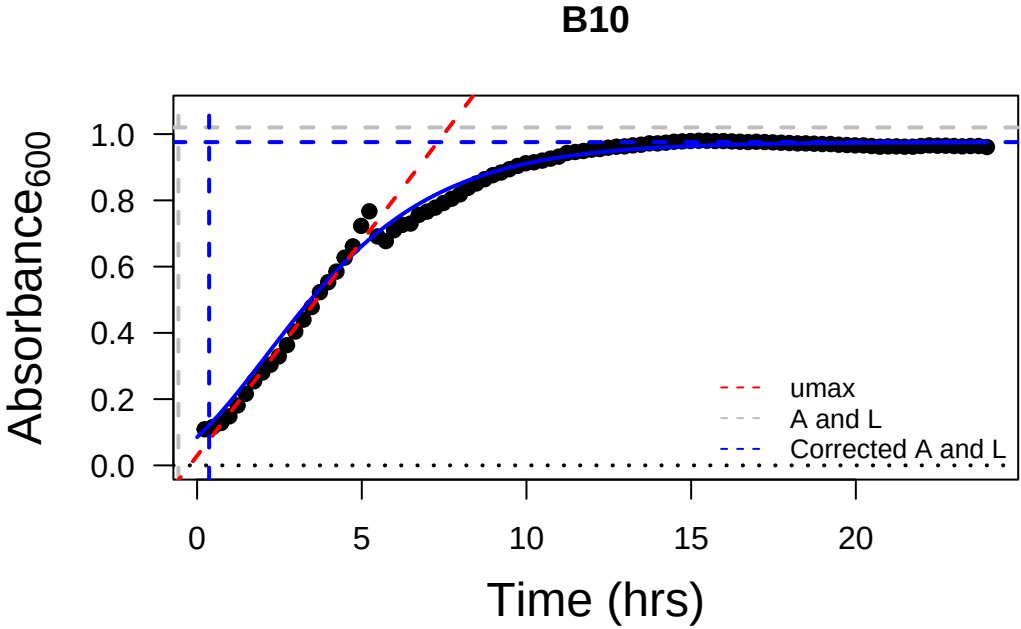

B5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

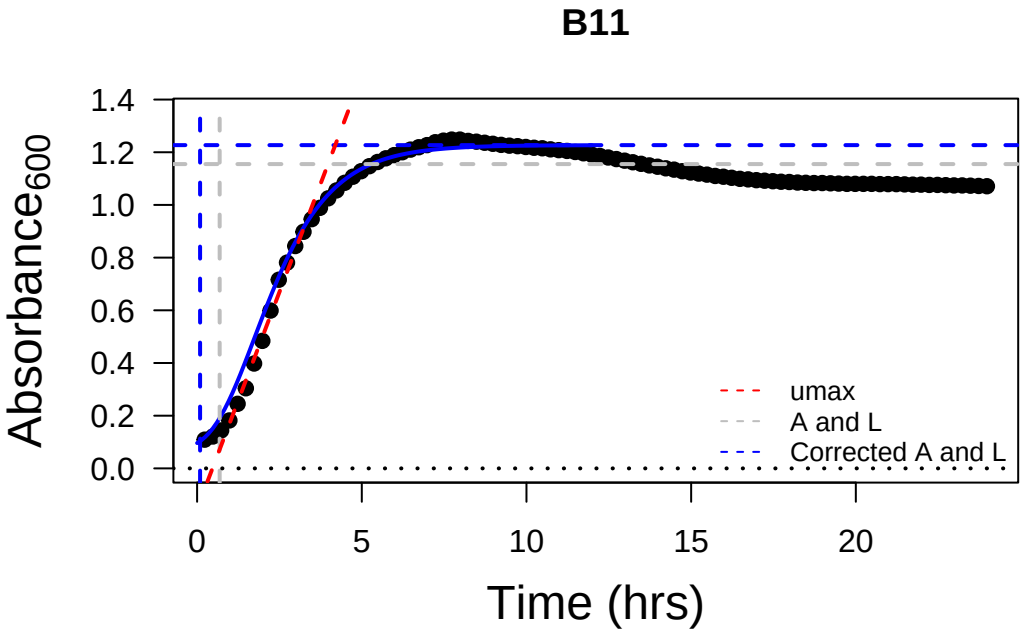

B6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

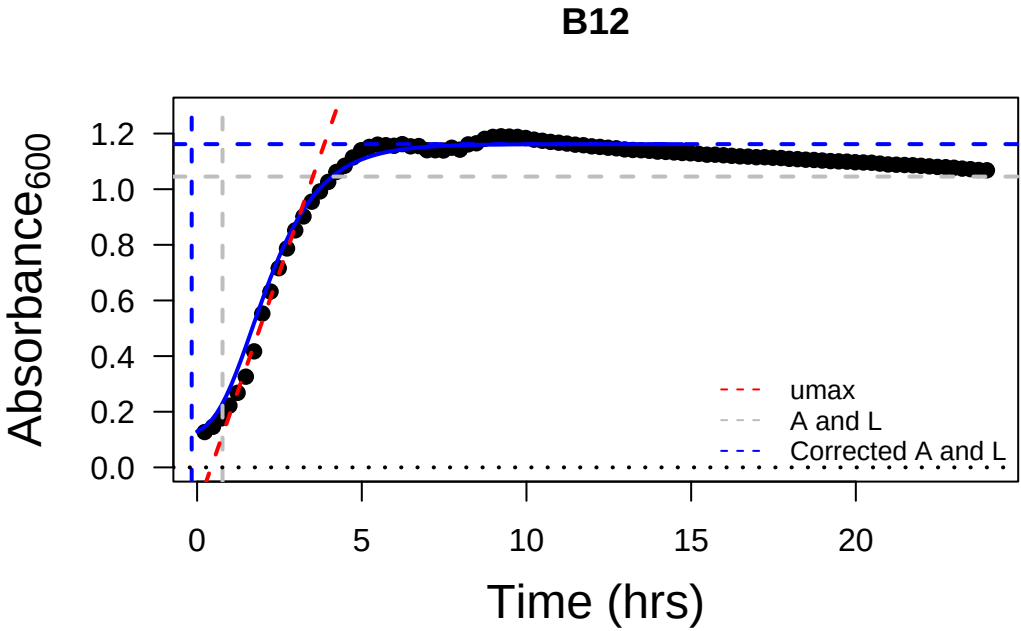

B7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

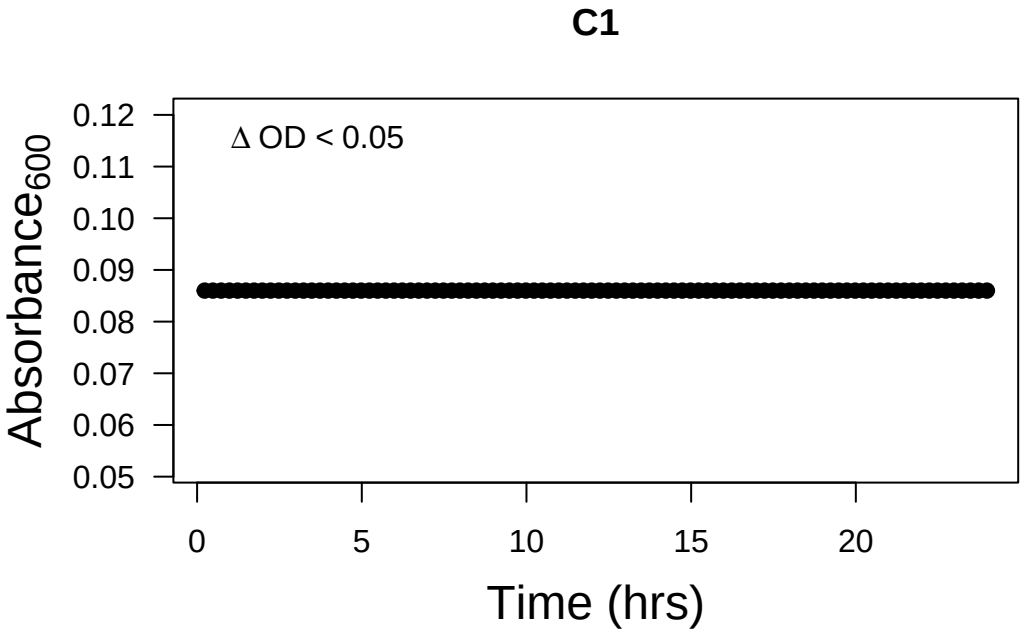

B8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

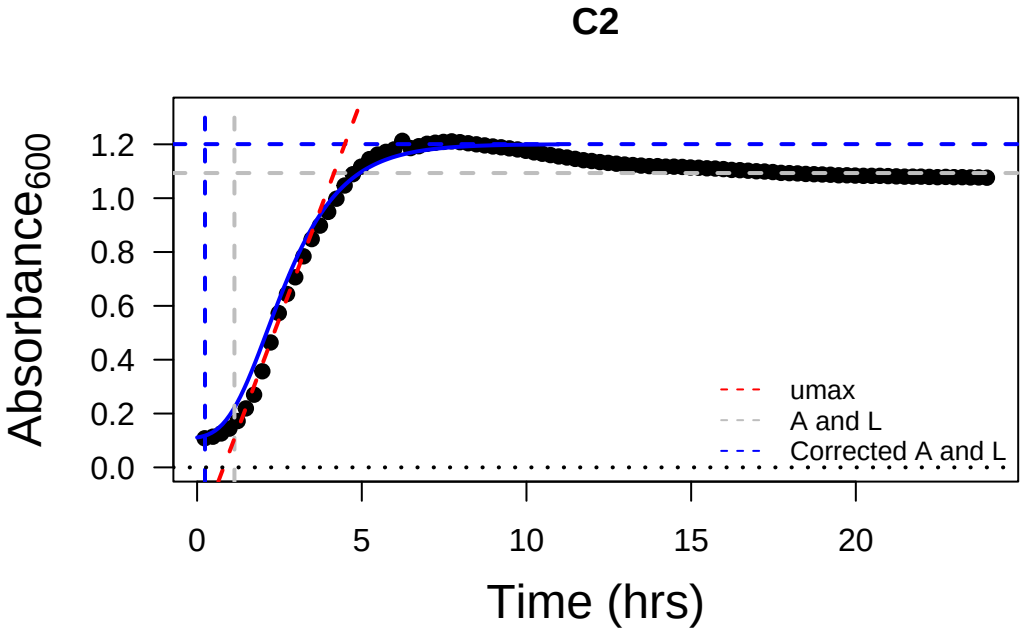

B9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

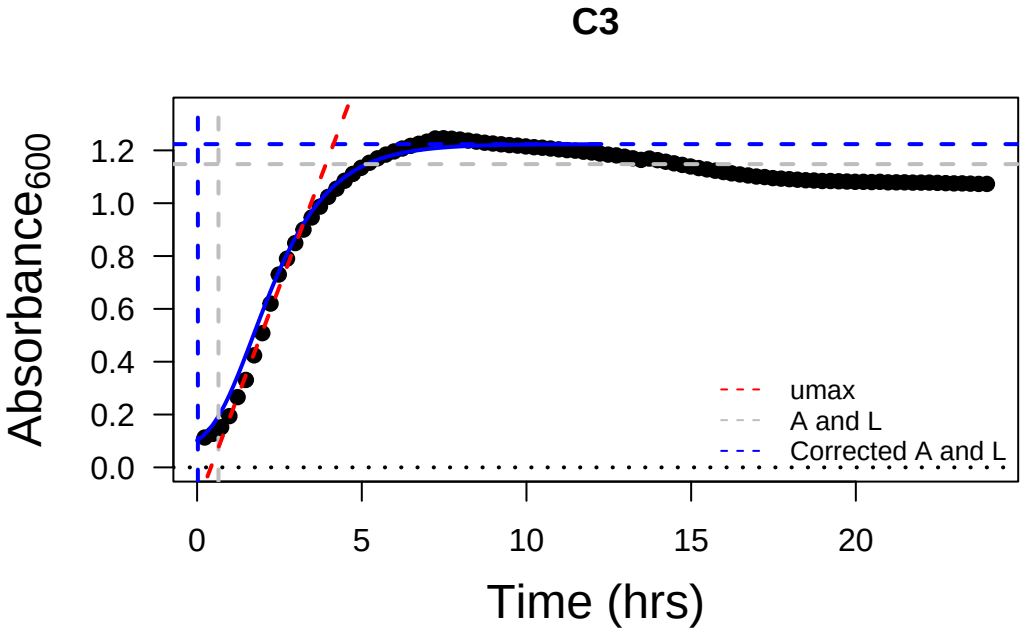

B10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

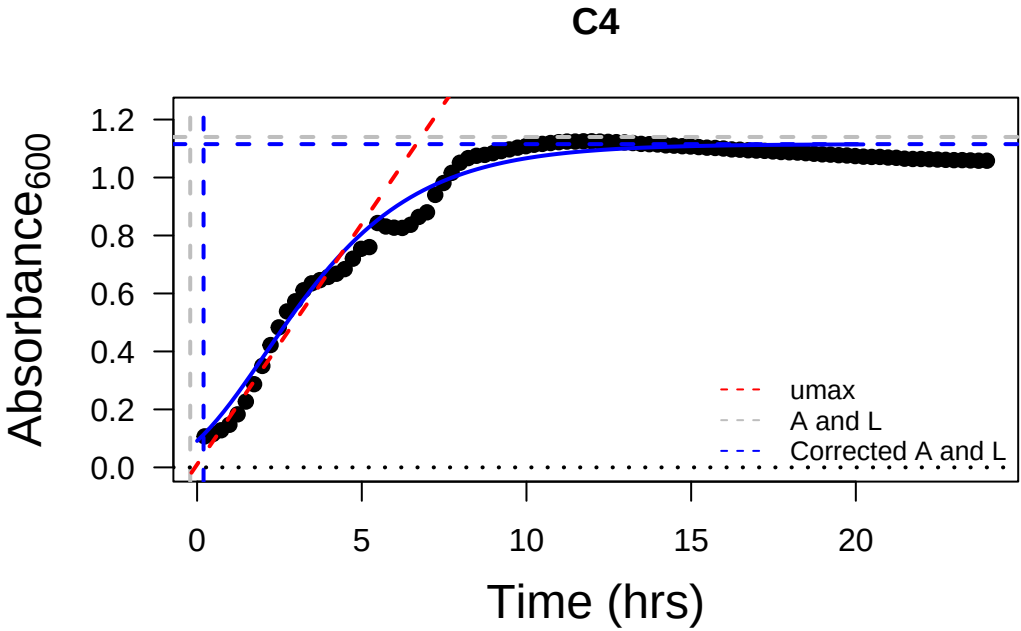

B11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

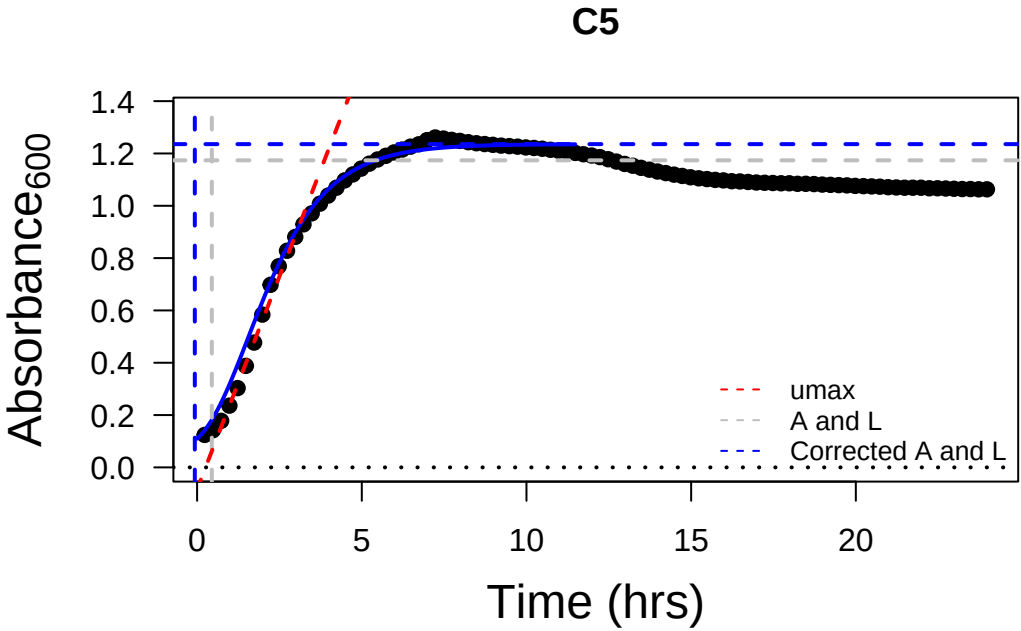

B12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

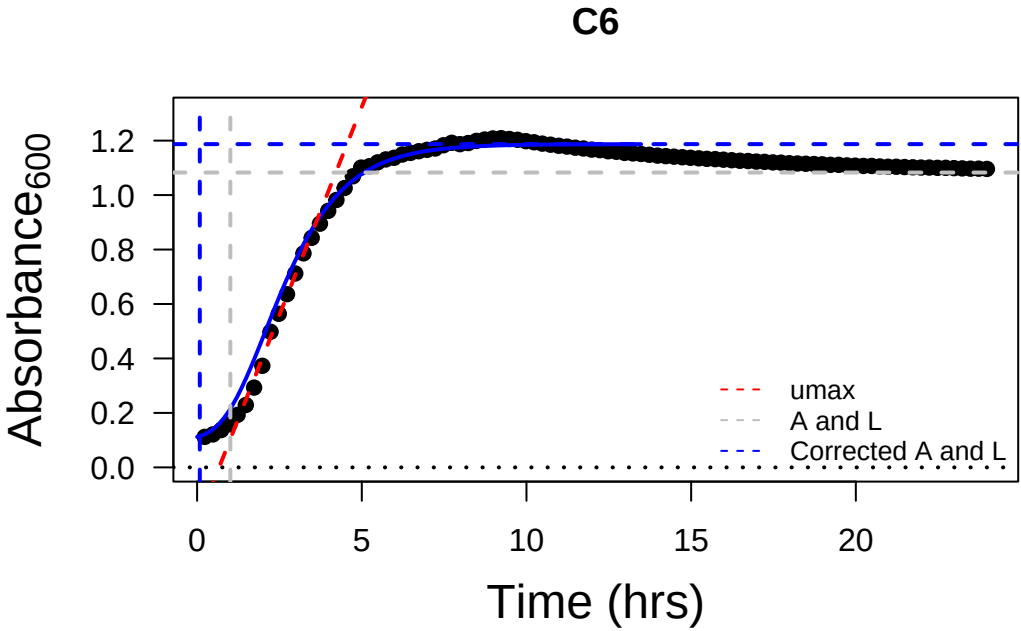

C1
Δ OD < 0.05
Absorbance600
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0
5
10
15
20
Time (hrs)

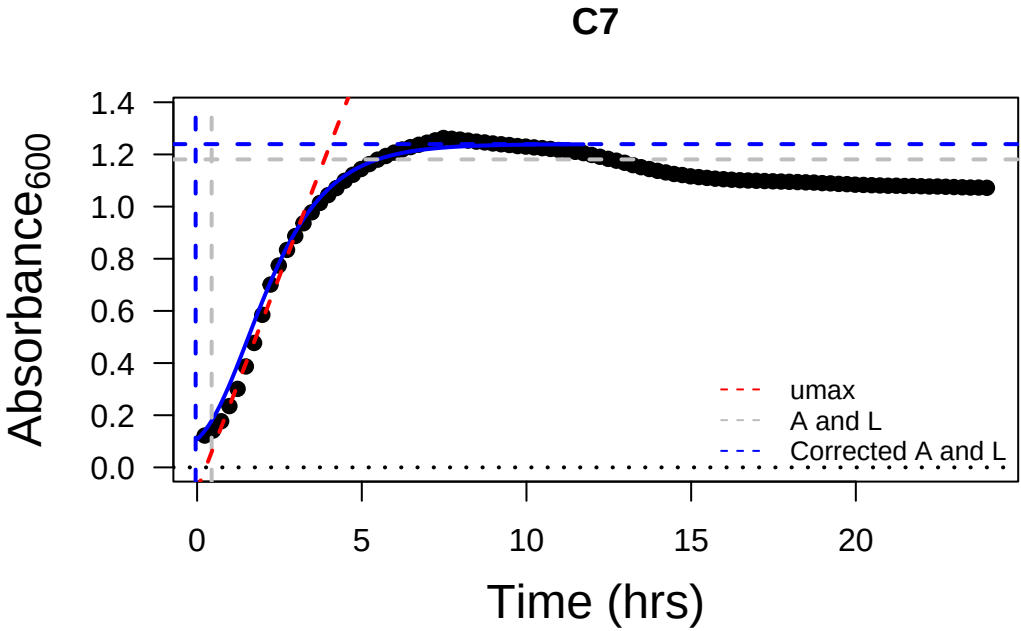

C2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

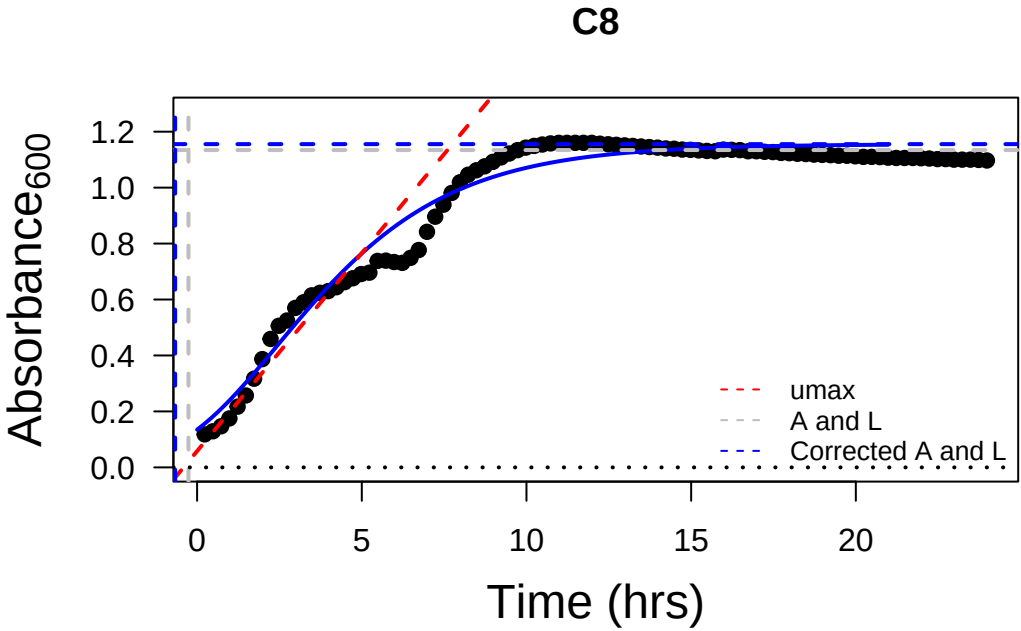

C3
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

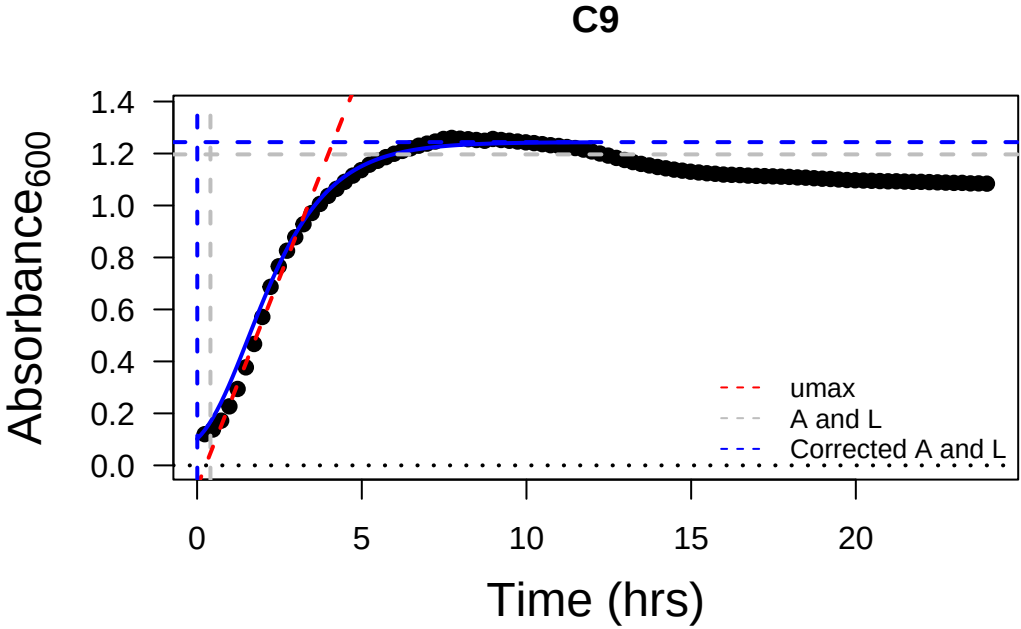

C4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

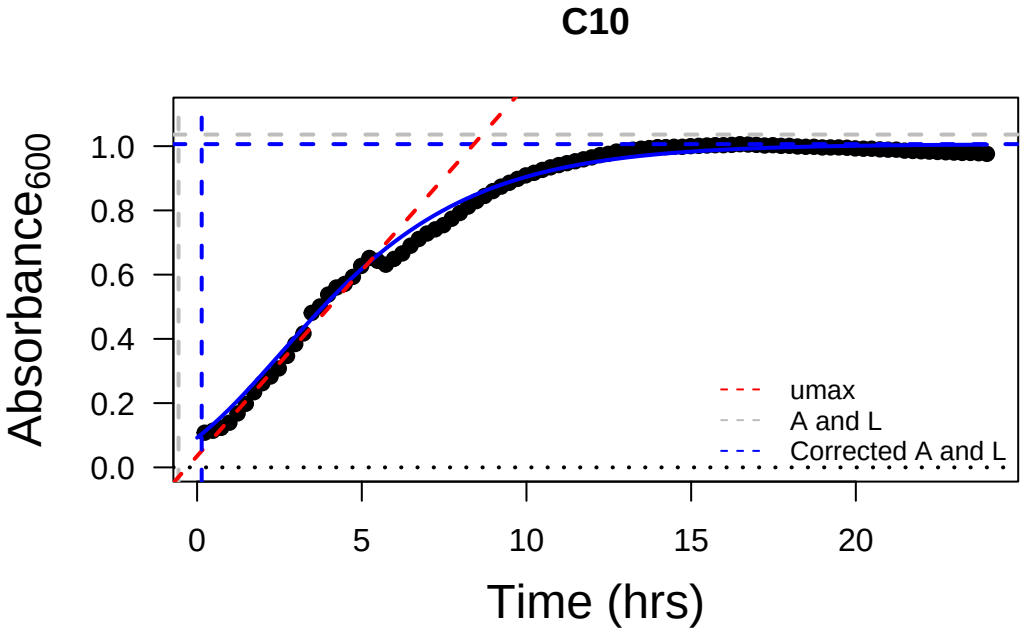

C5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

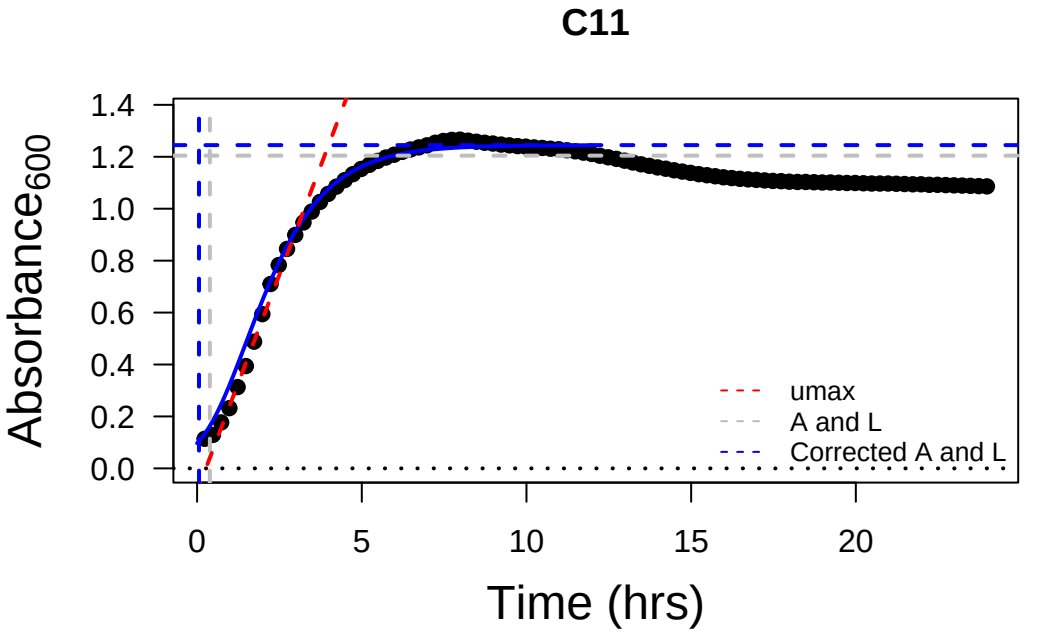

C6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

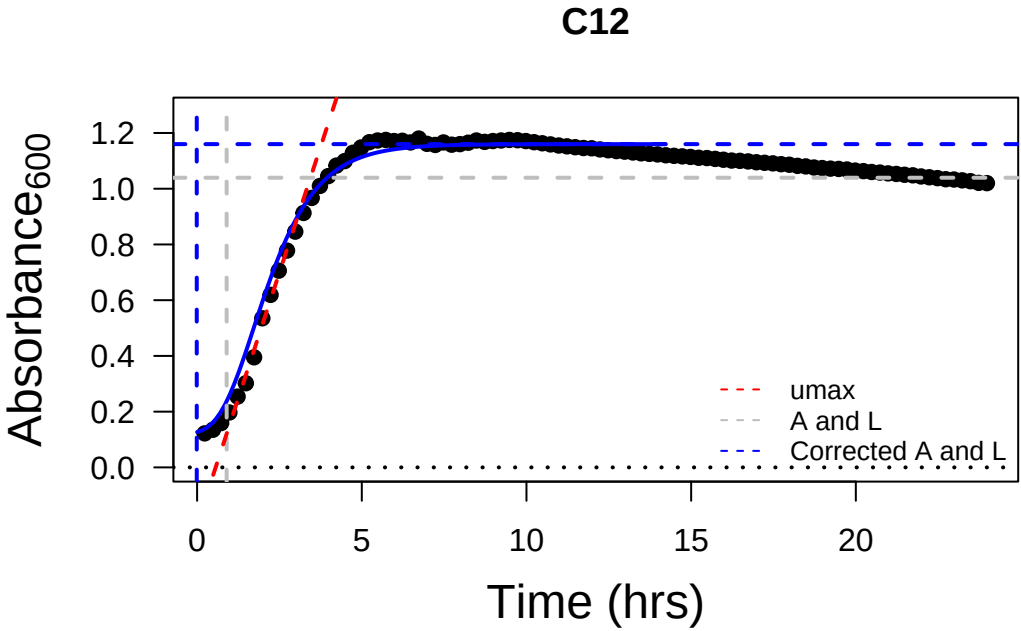

C7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

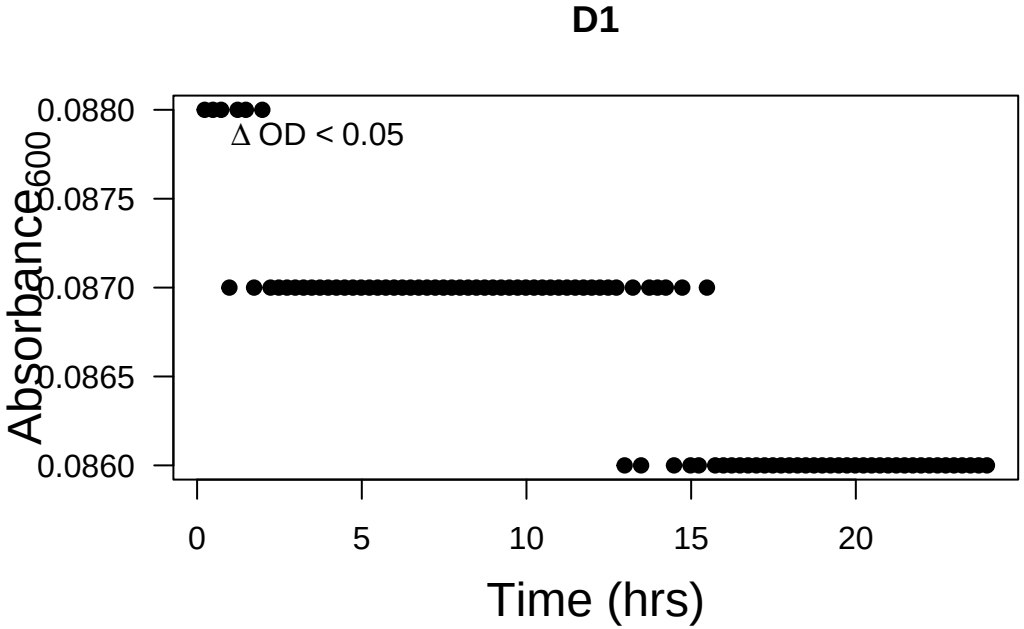

C8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

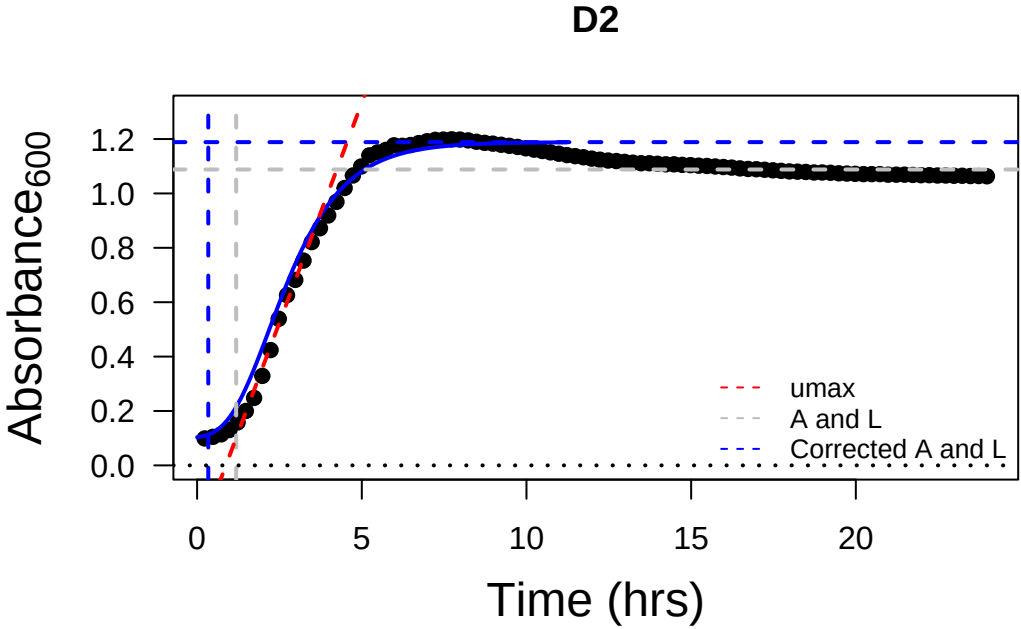

C9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

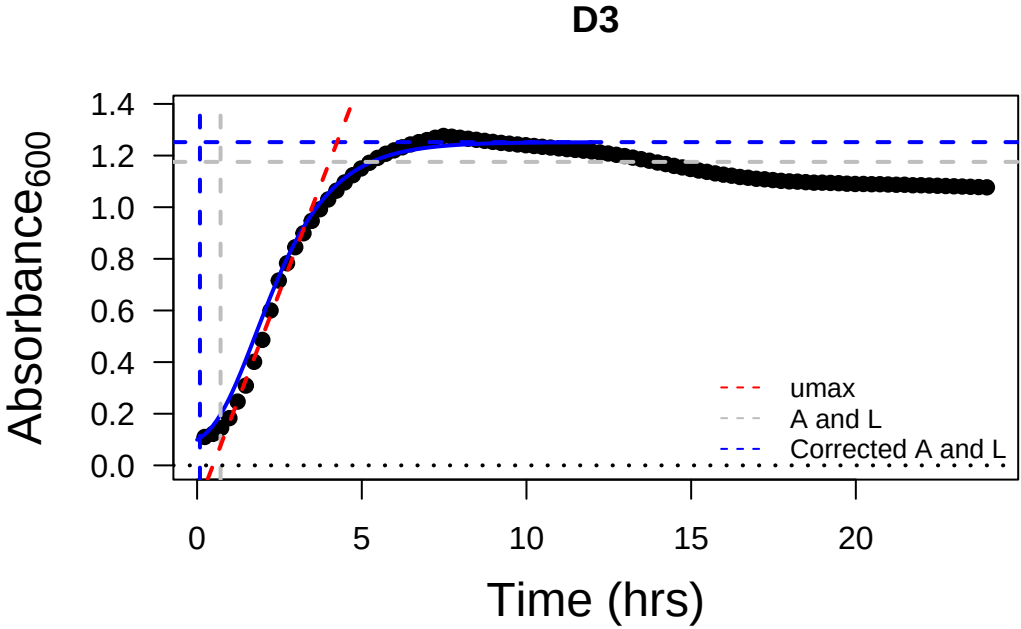

C10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

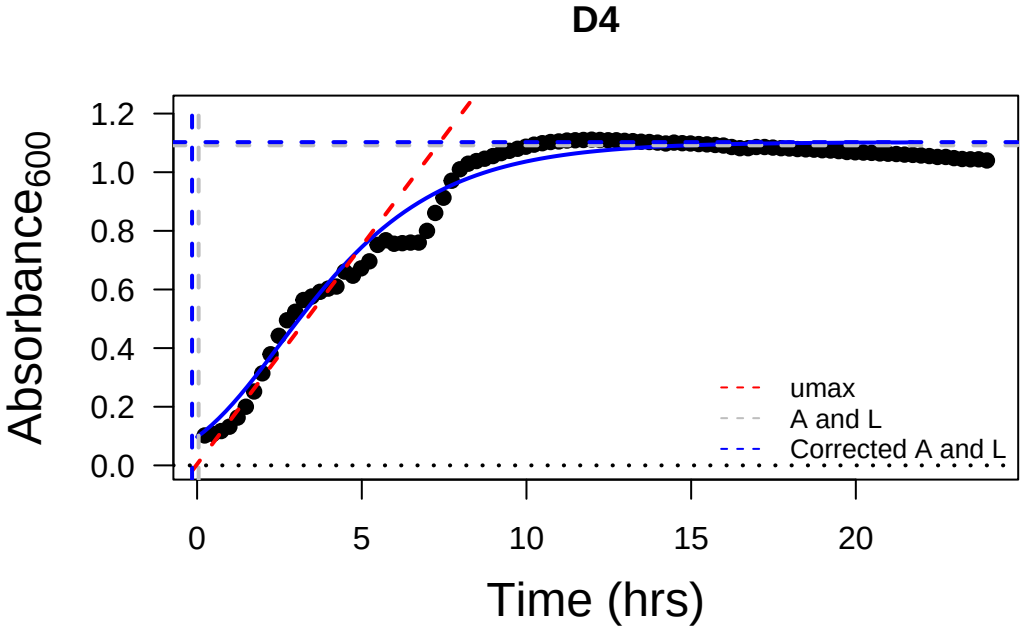

C11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

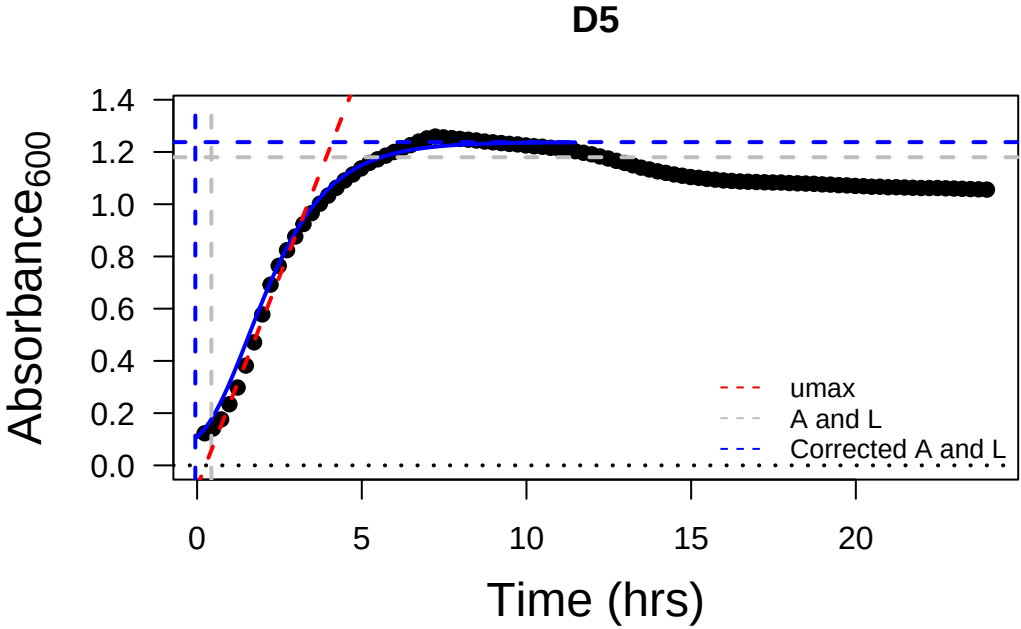

C12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

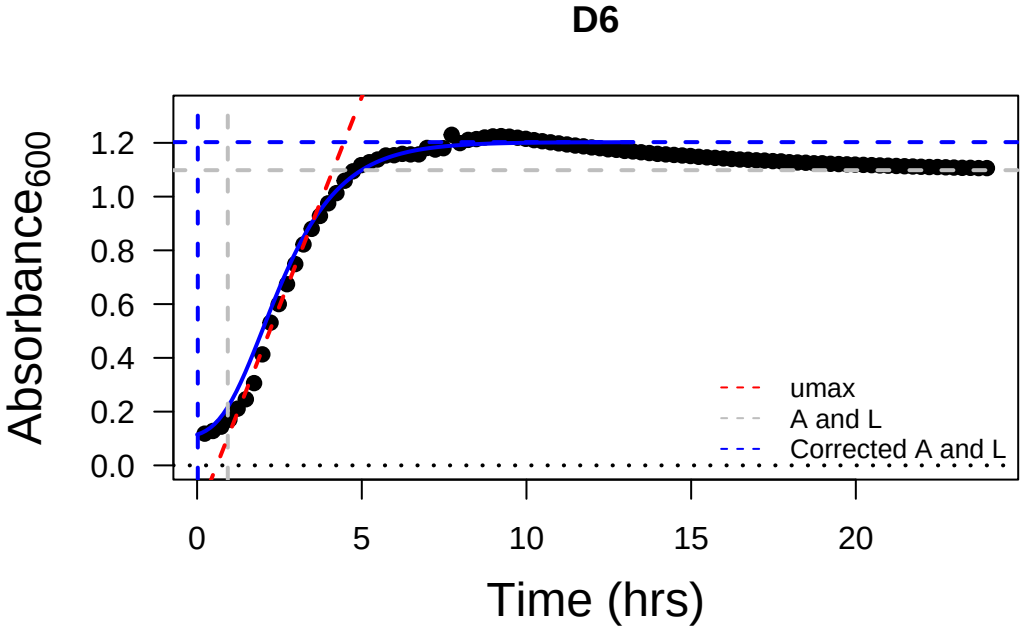

D1
Absorbance600
0.0880
0.0875
0.0870
0.0865
0.0860
Δ OD < 0.05
0
5
10
15
20
Time (hrs)

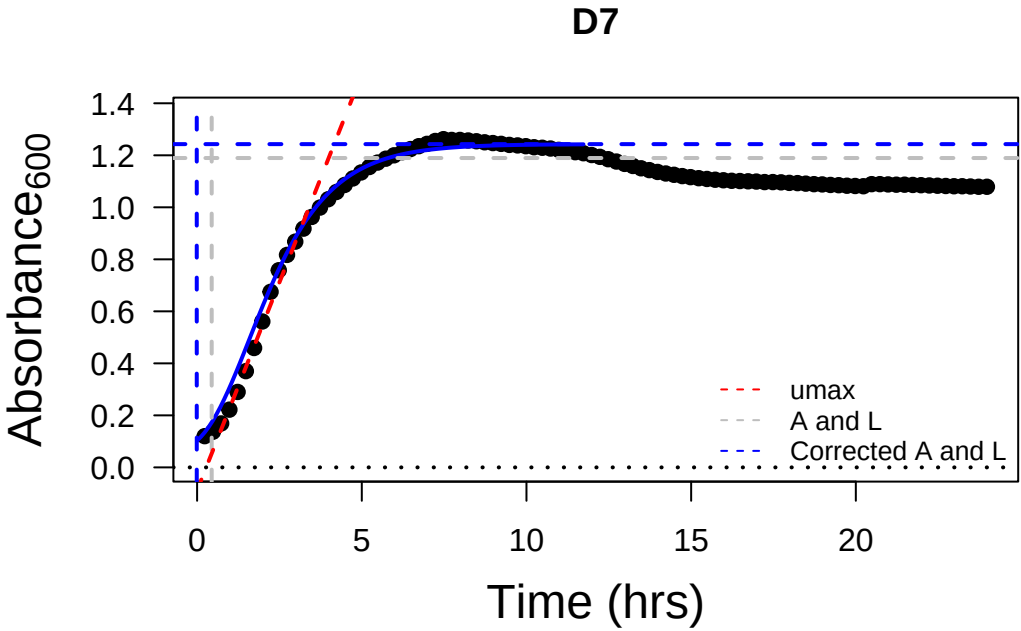

D2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

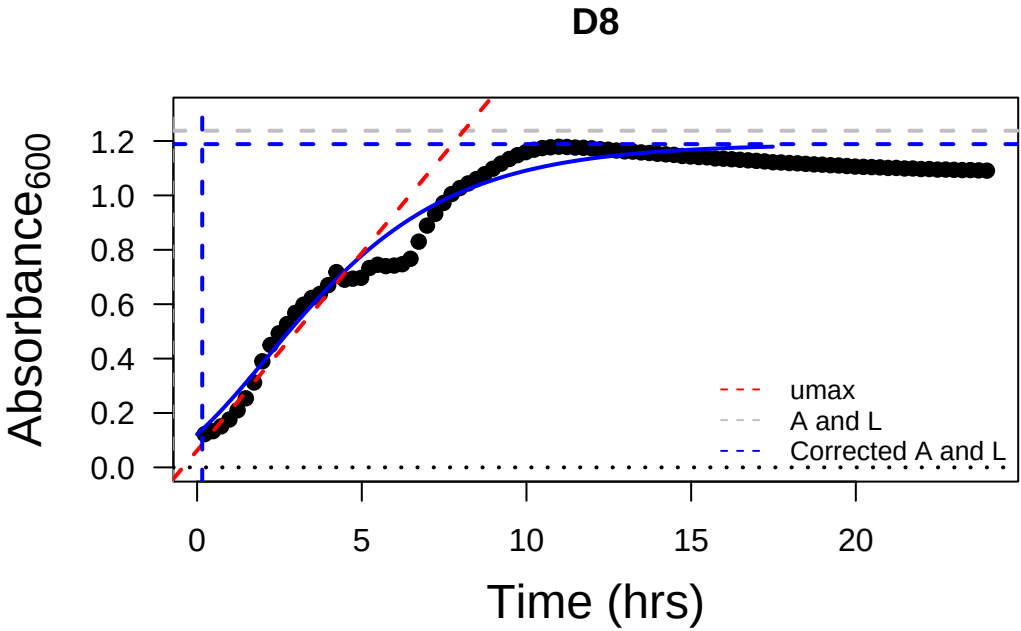

D3
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

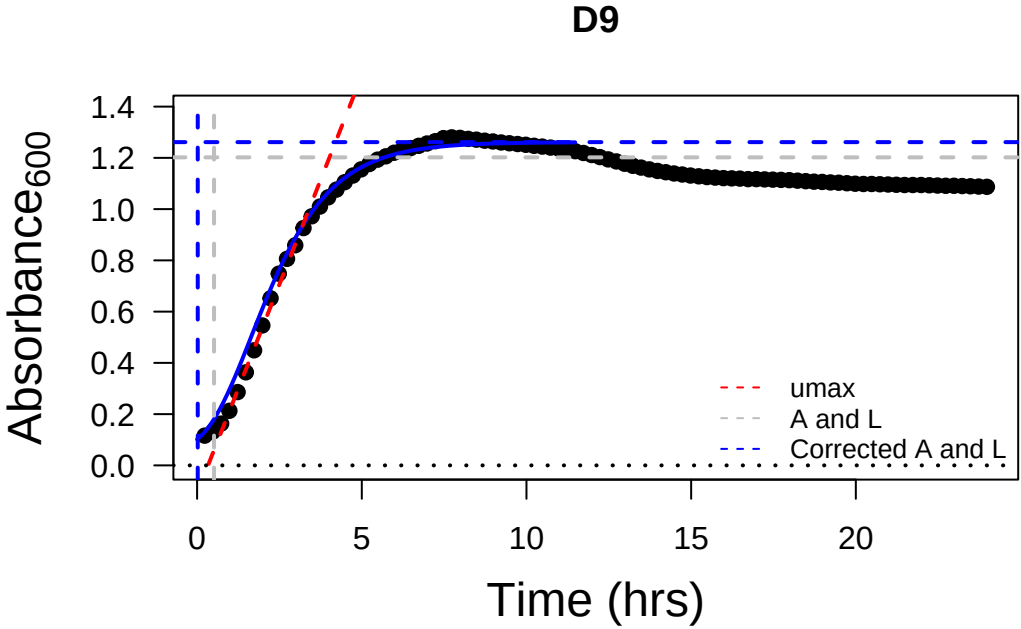

D4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

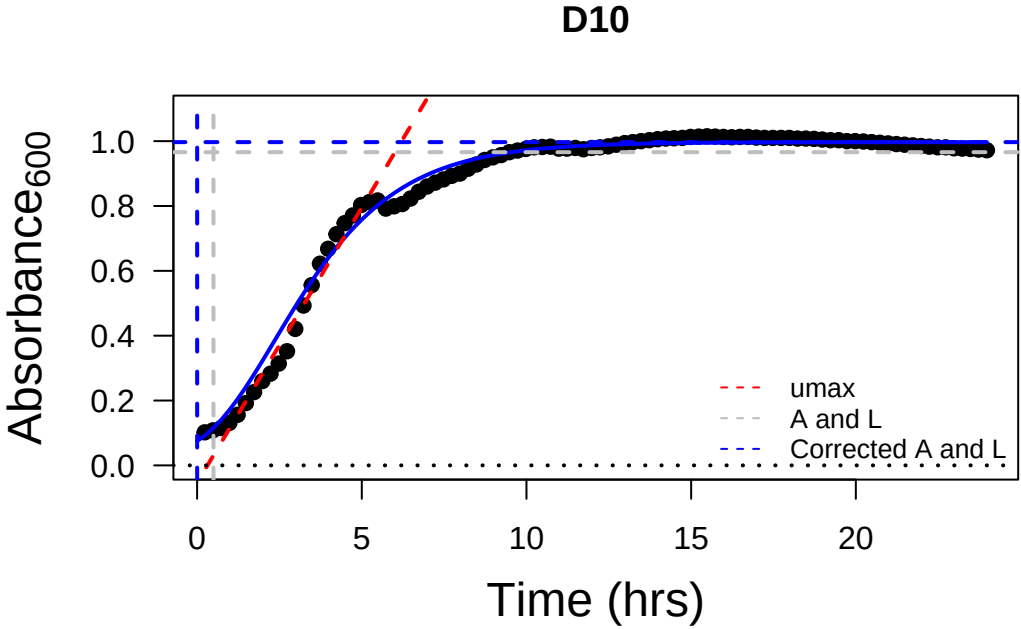

D5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

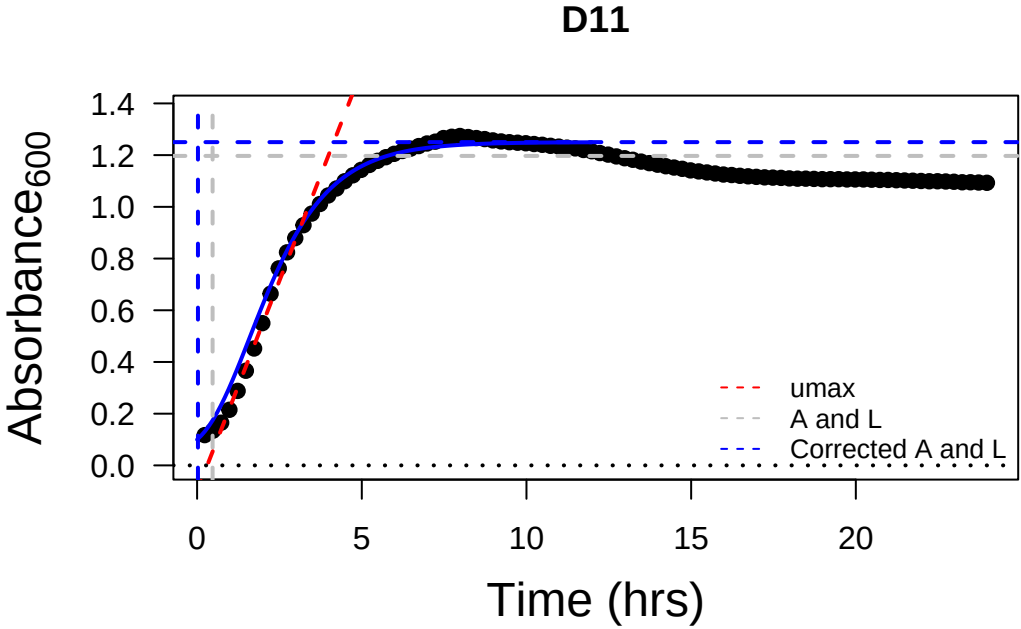

D6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

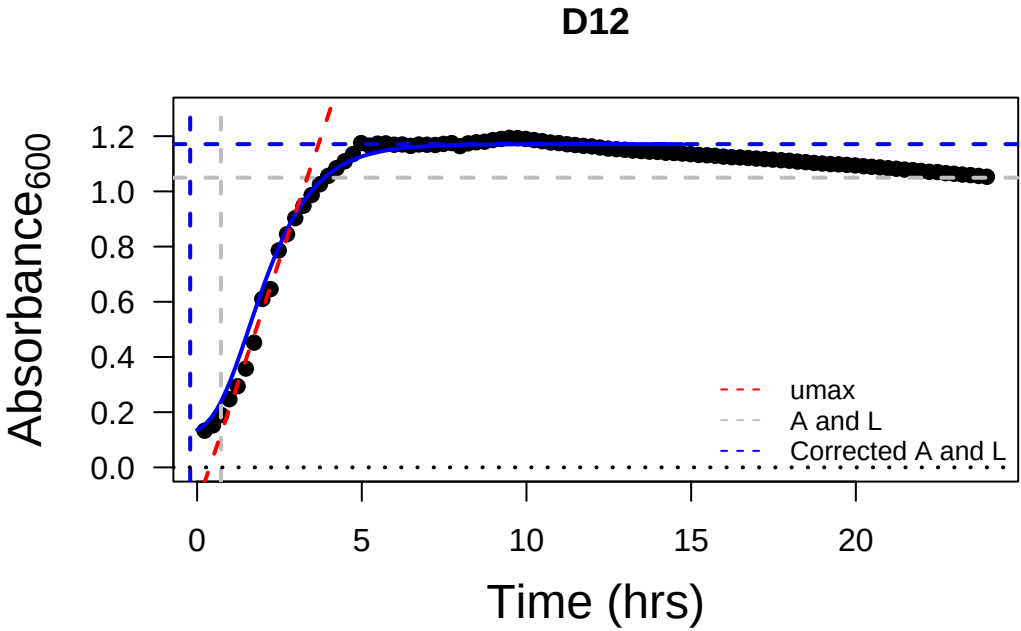

D7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

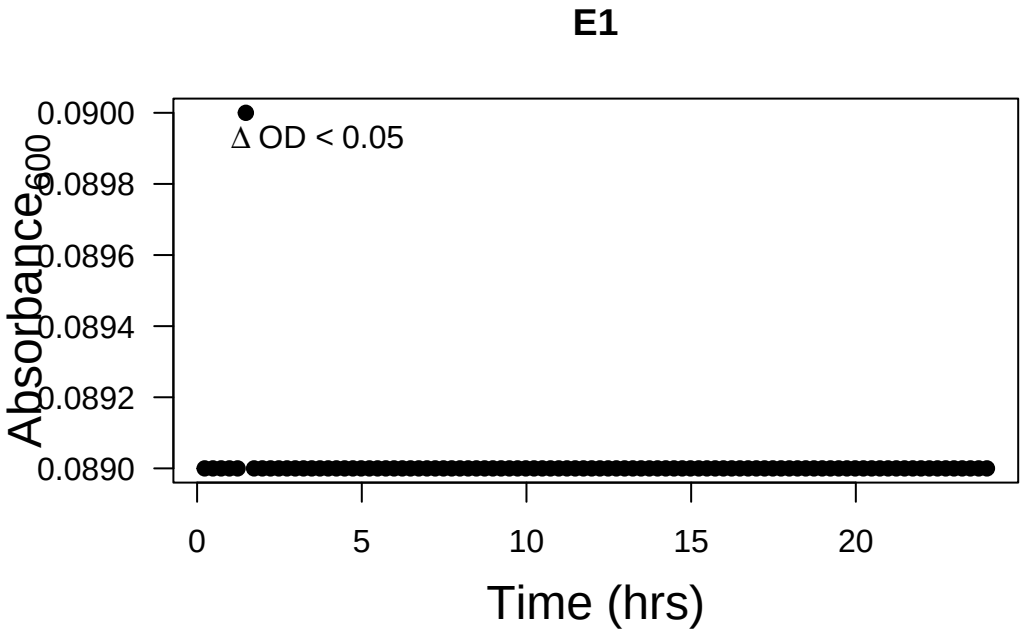

D8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

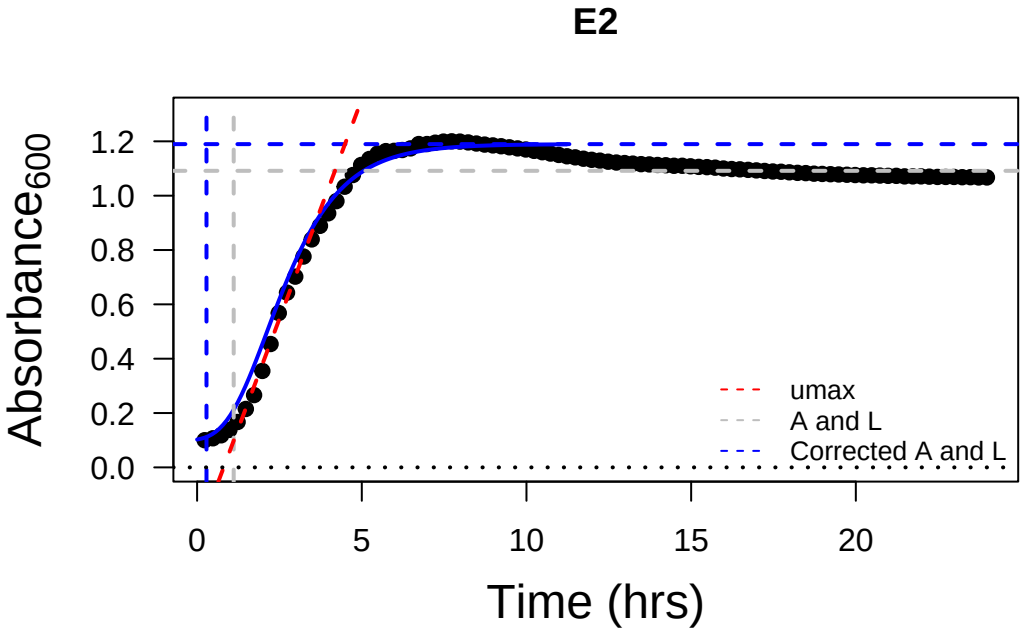

D9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

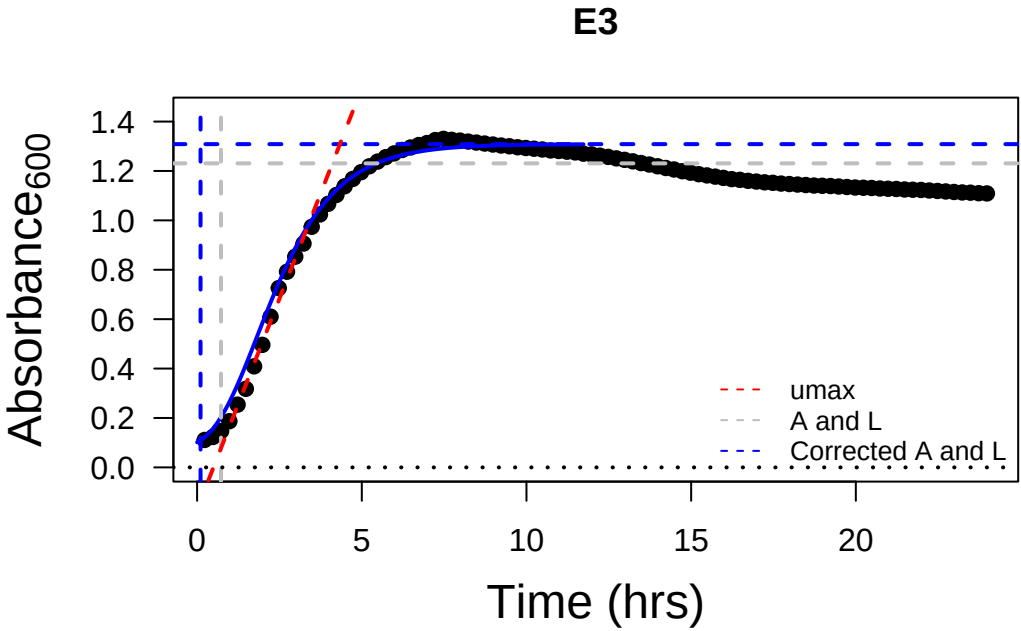

D10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

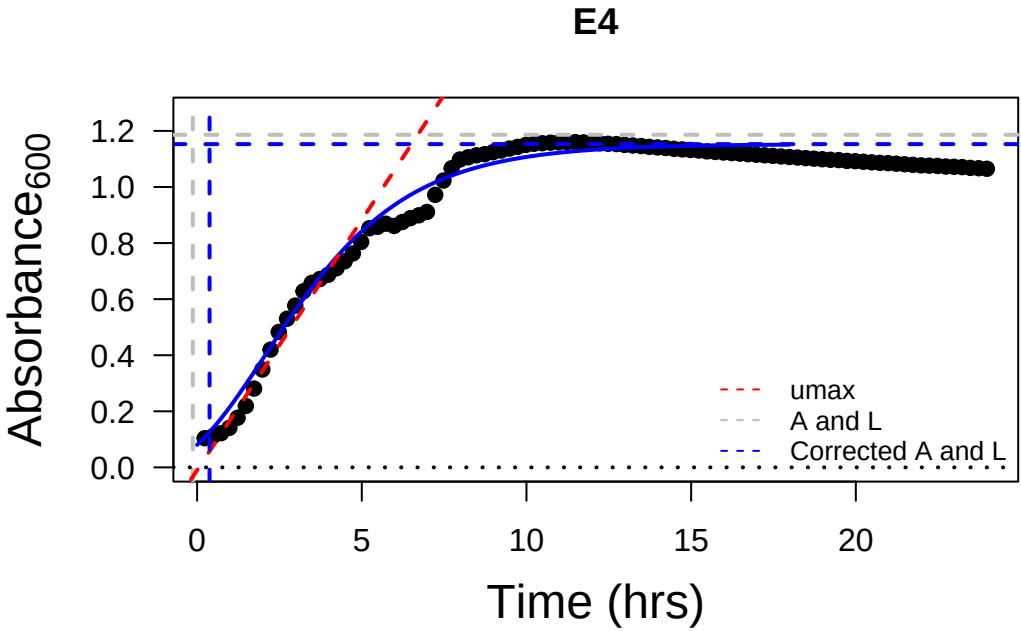

D11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

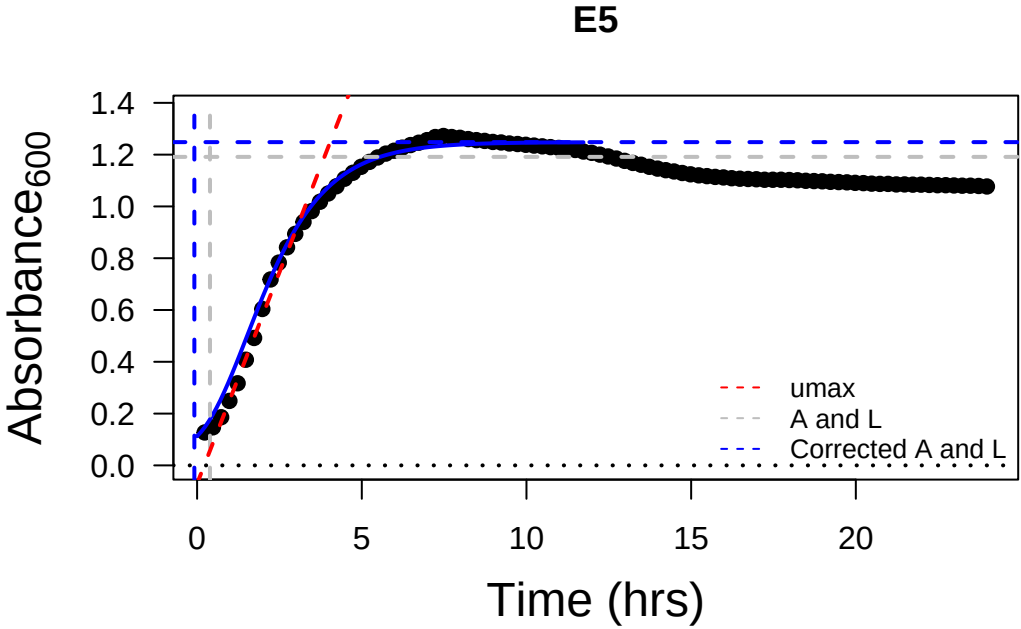

D12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

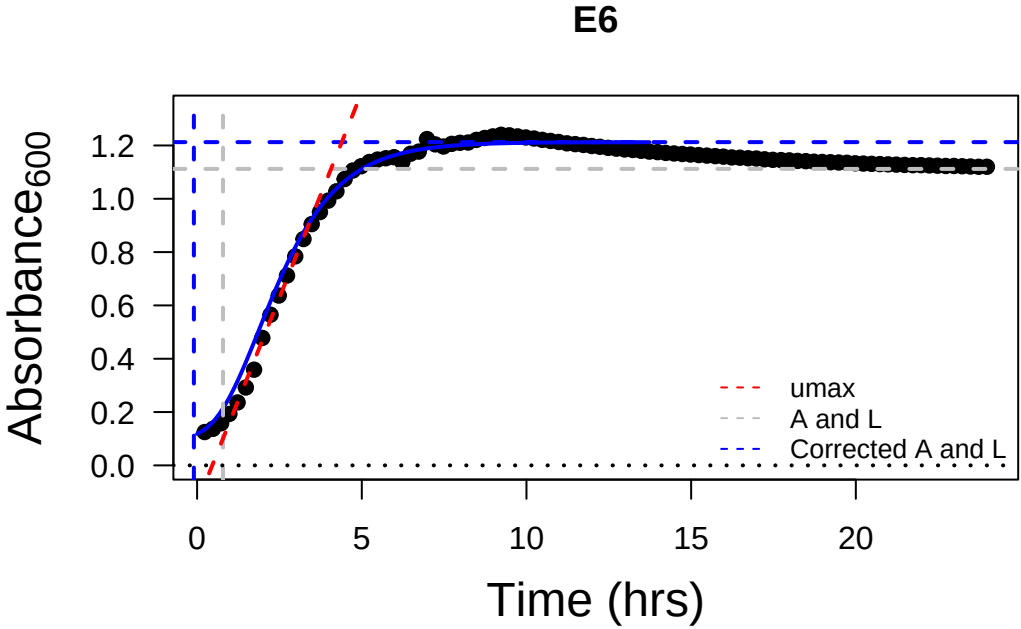

E1
Absorbance600
0.0900
0.0898
0.0896
0.0894
0.0892
0.0890
Δ OD < 0.05
0
5
10
15
20
Time (hrs)

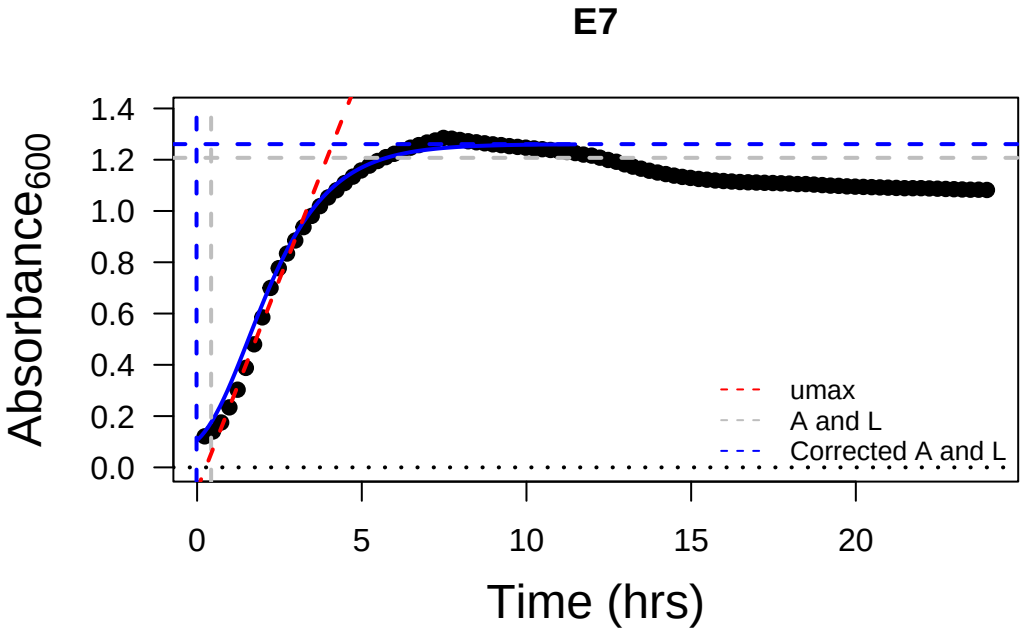

E2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

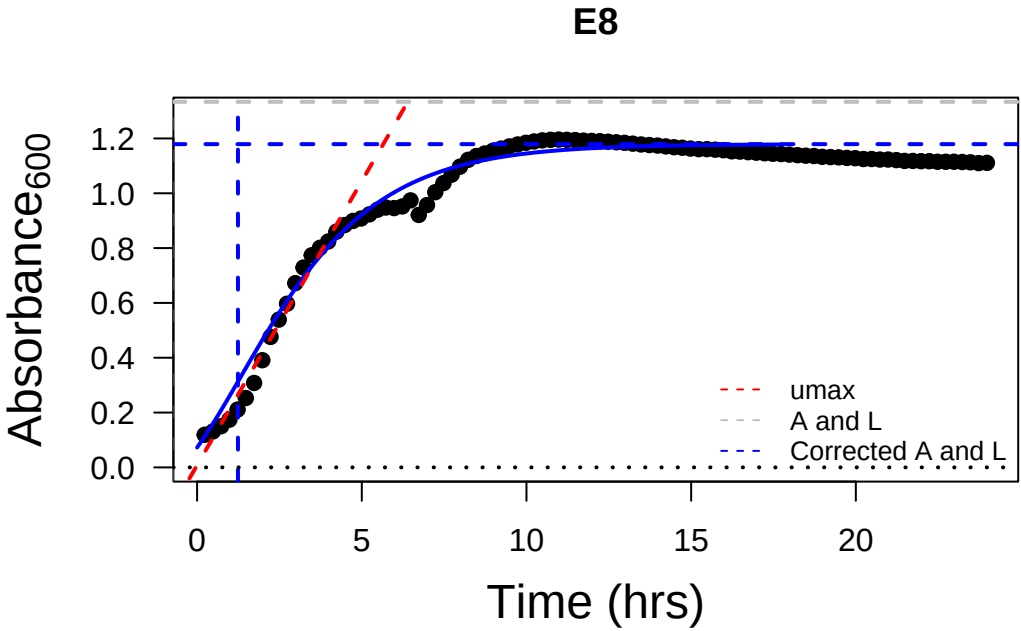

E3
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

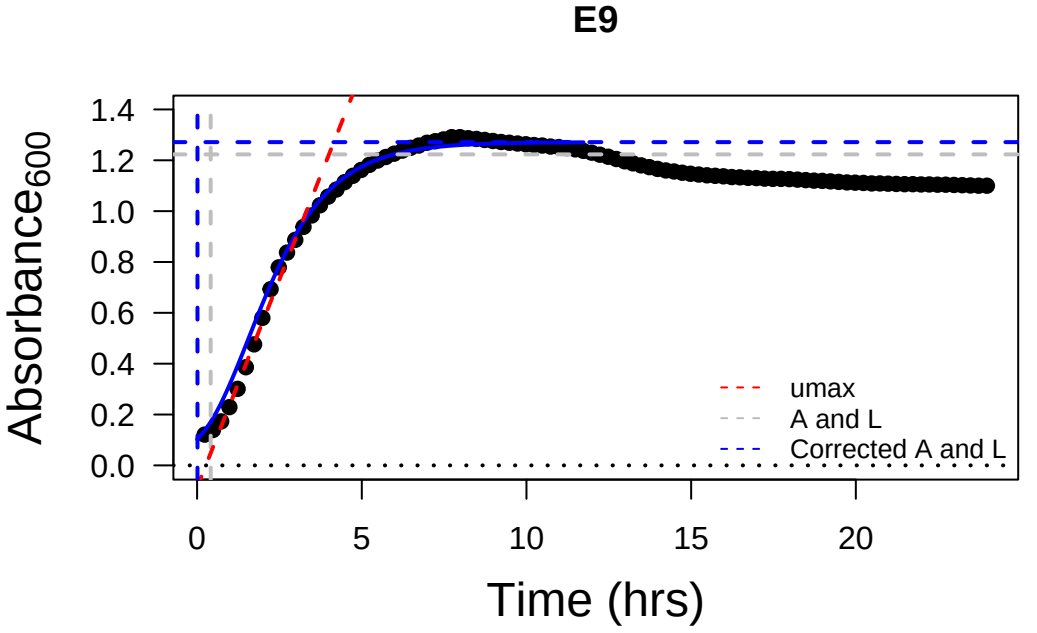

E4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

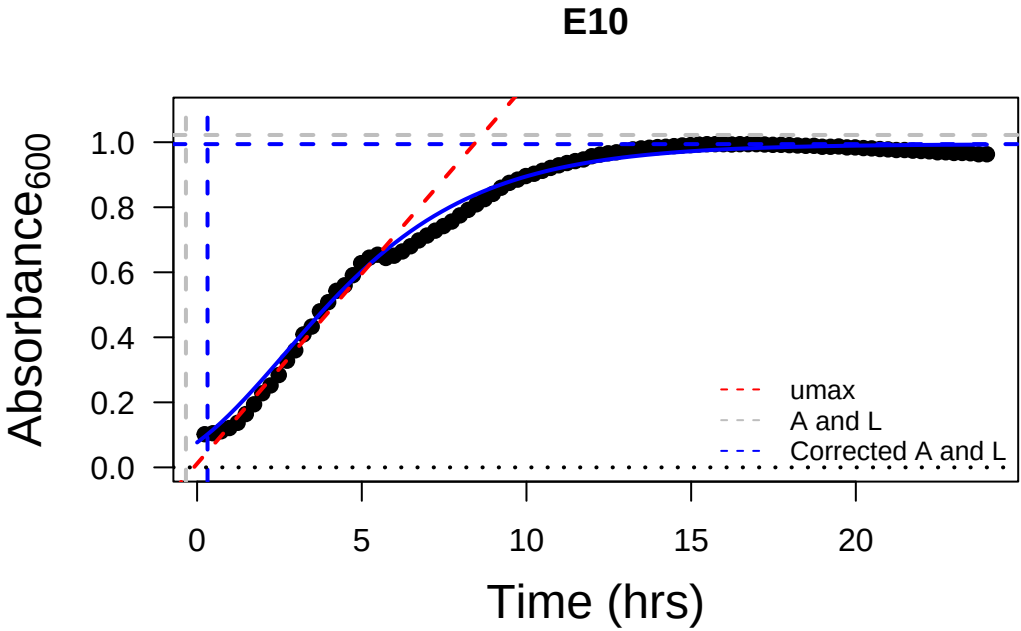

E5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

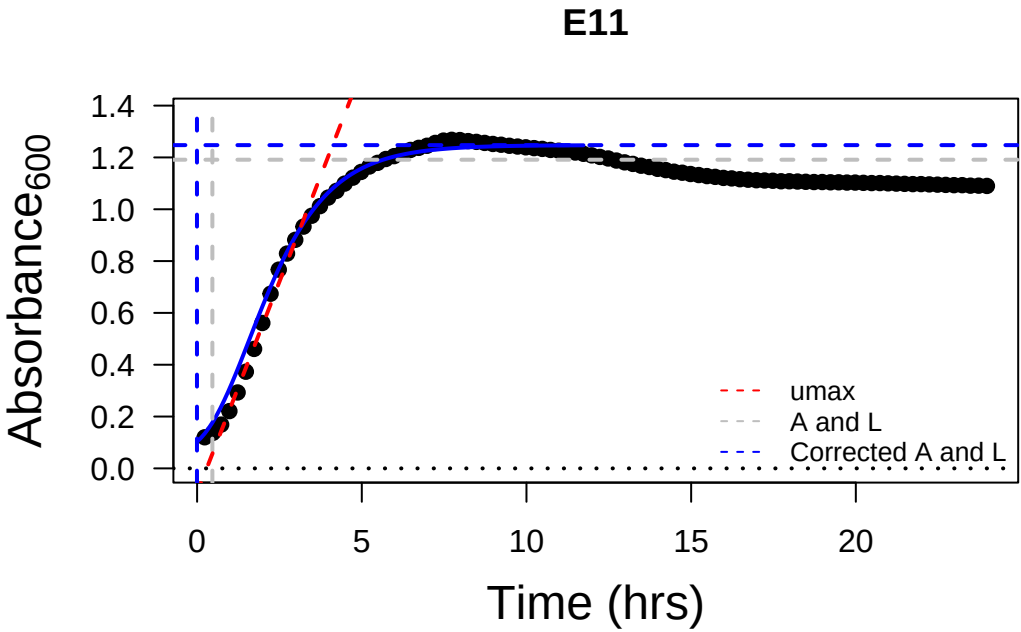

E6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

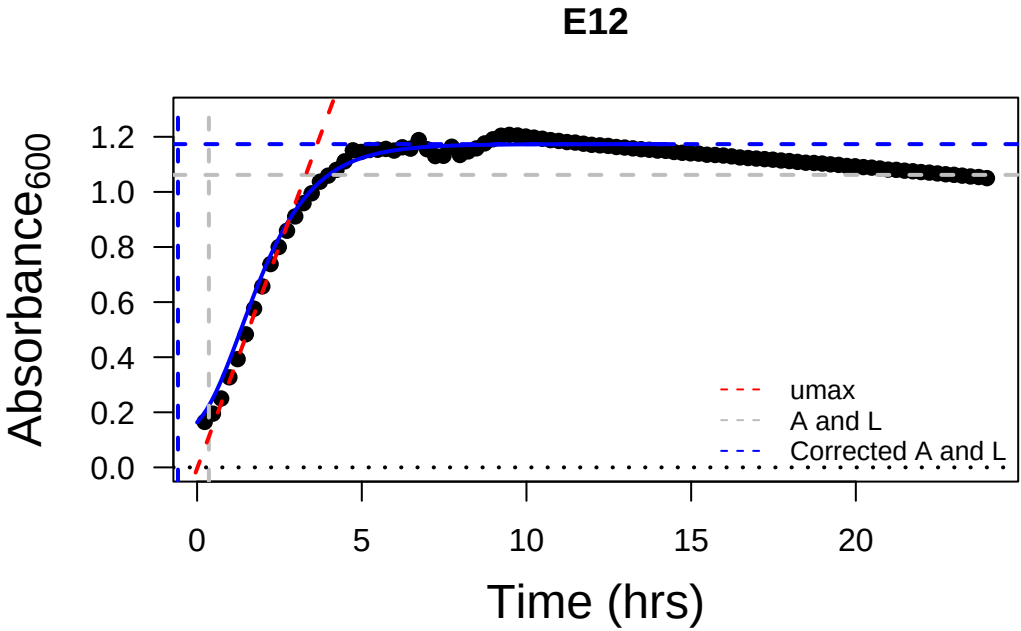

E7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

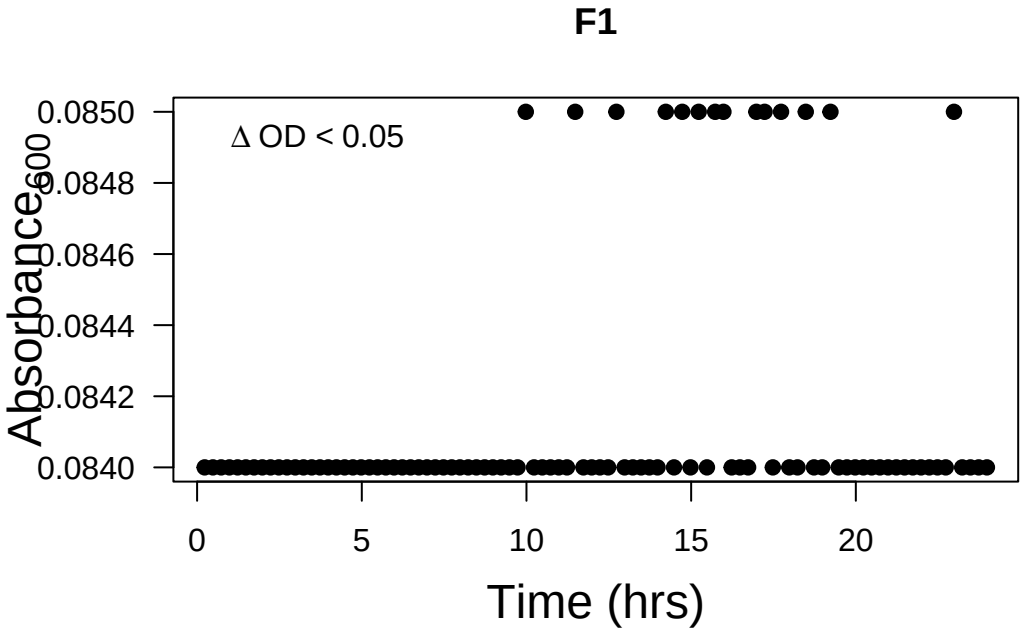

E8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

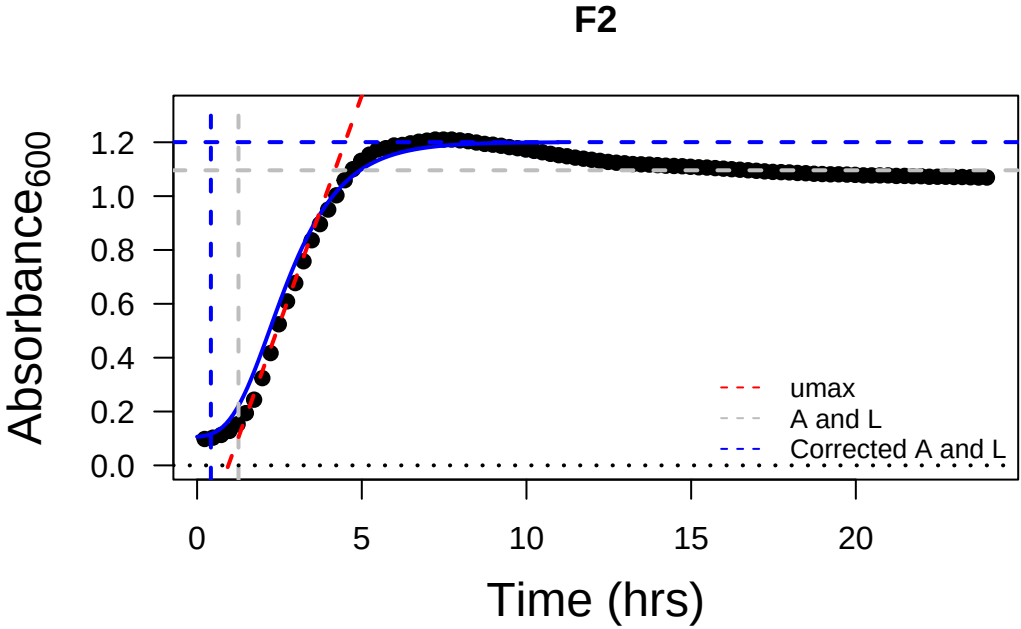

E9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

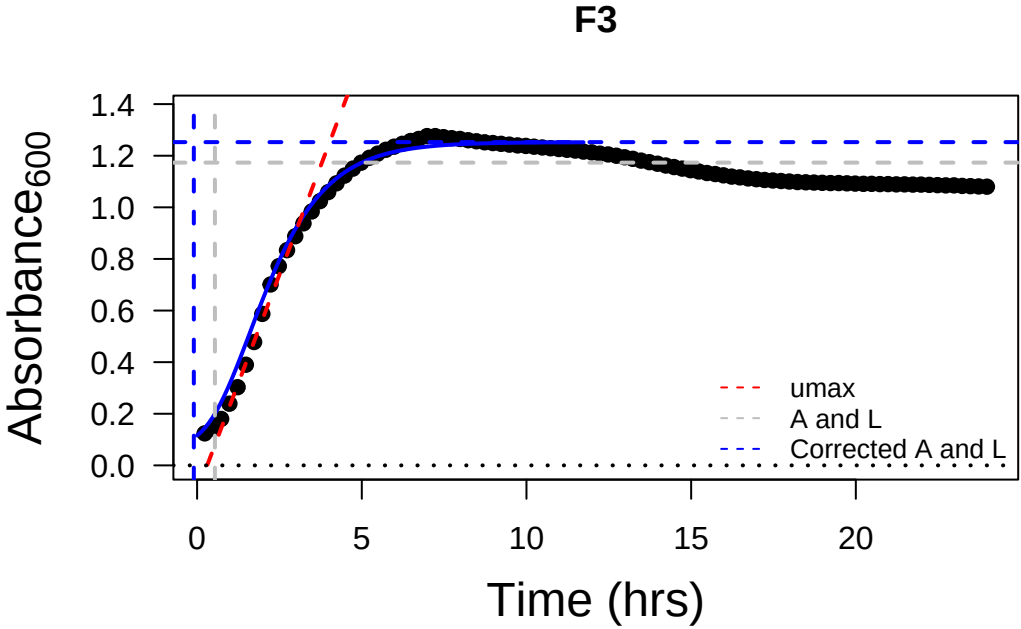

E10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

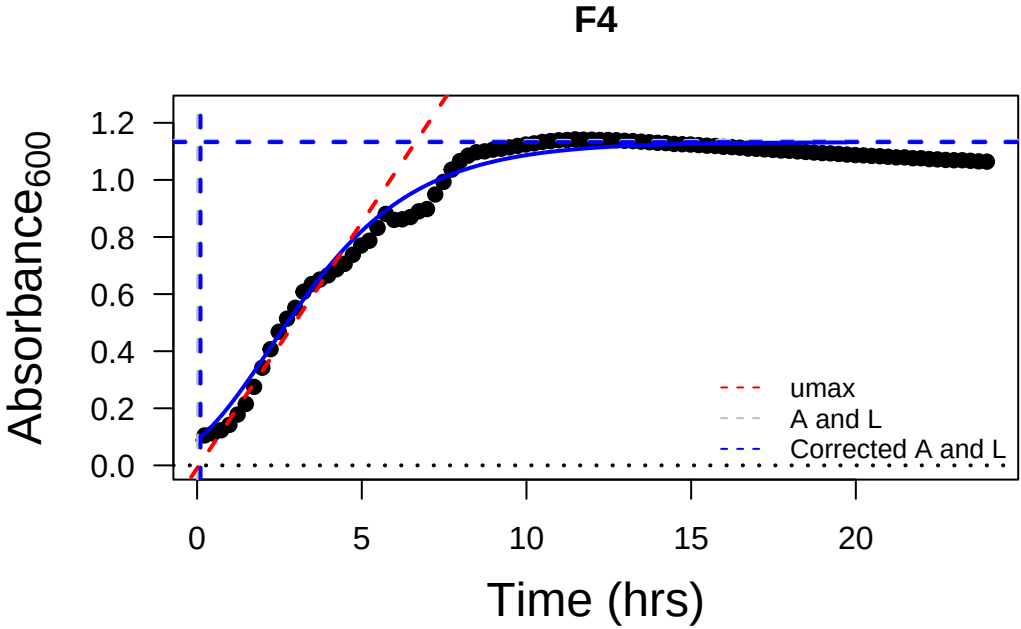

E11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

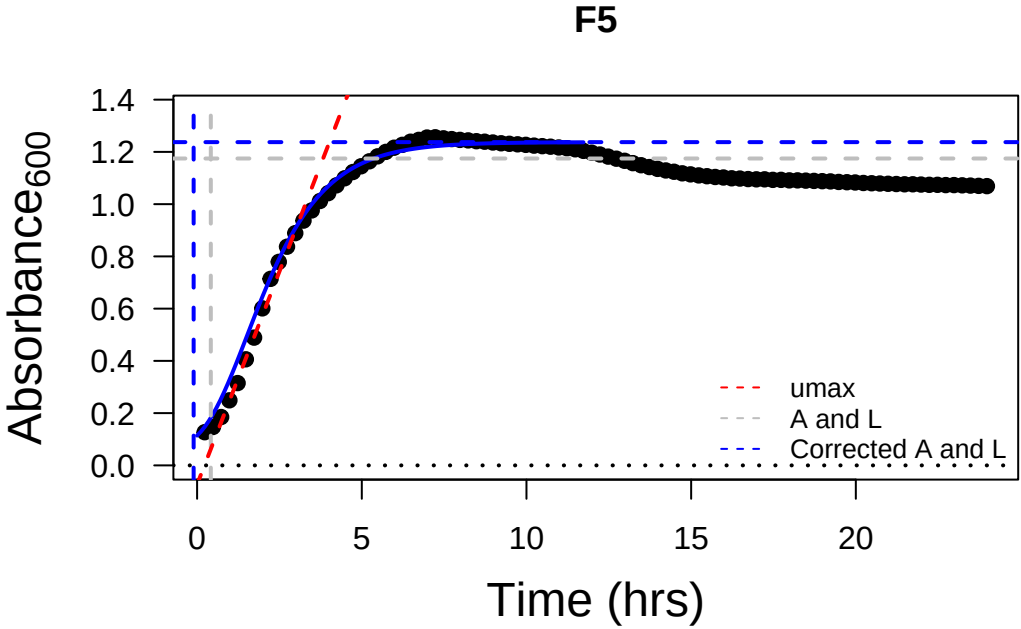

E12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

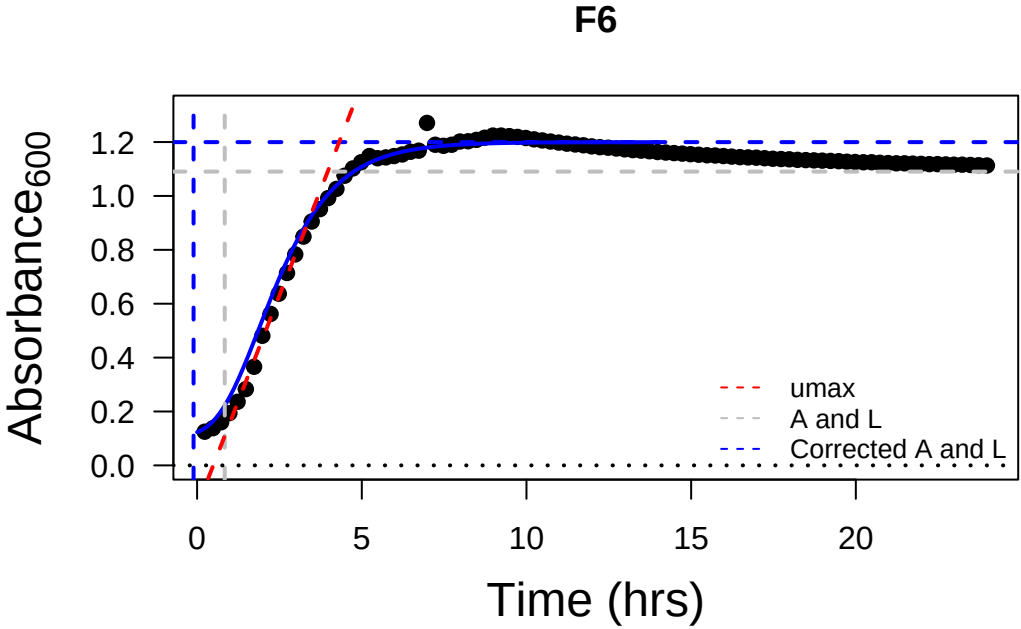

F1
Δ OD < 0.05
Absorbance600
0.0850
0.0848
0.0846
0.0844
0.0842
0.0840
0
5
10
15
20
Time (hrs)

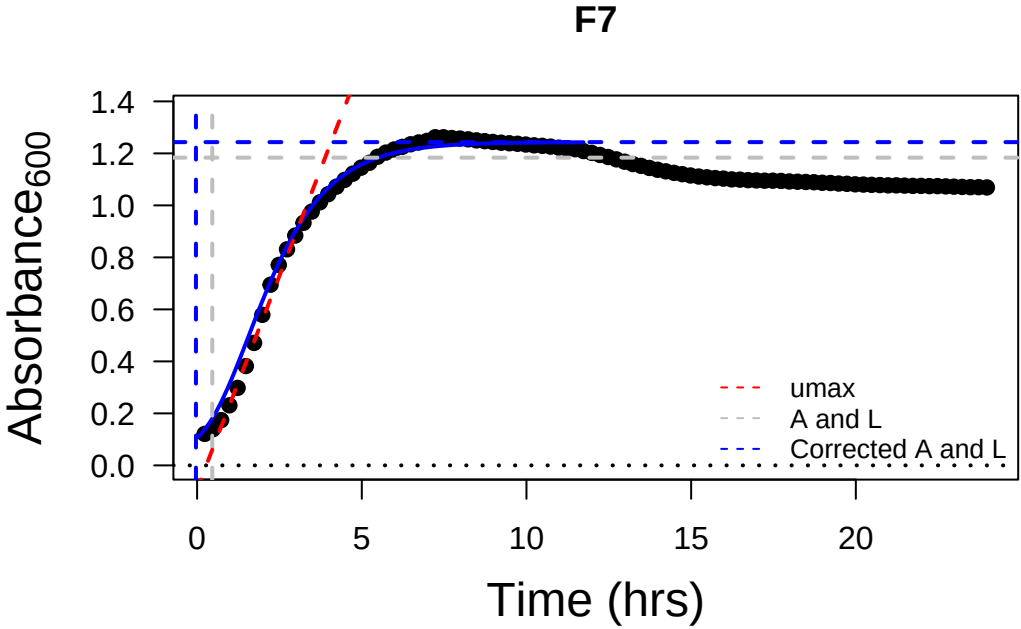

F2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

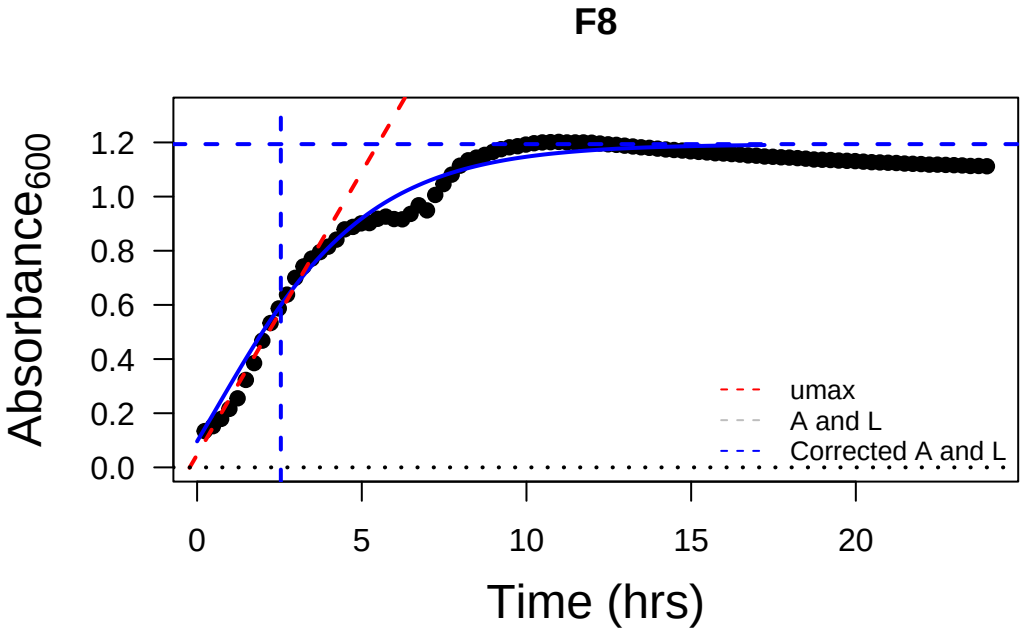

F3
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

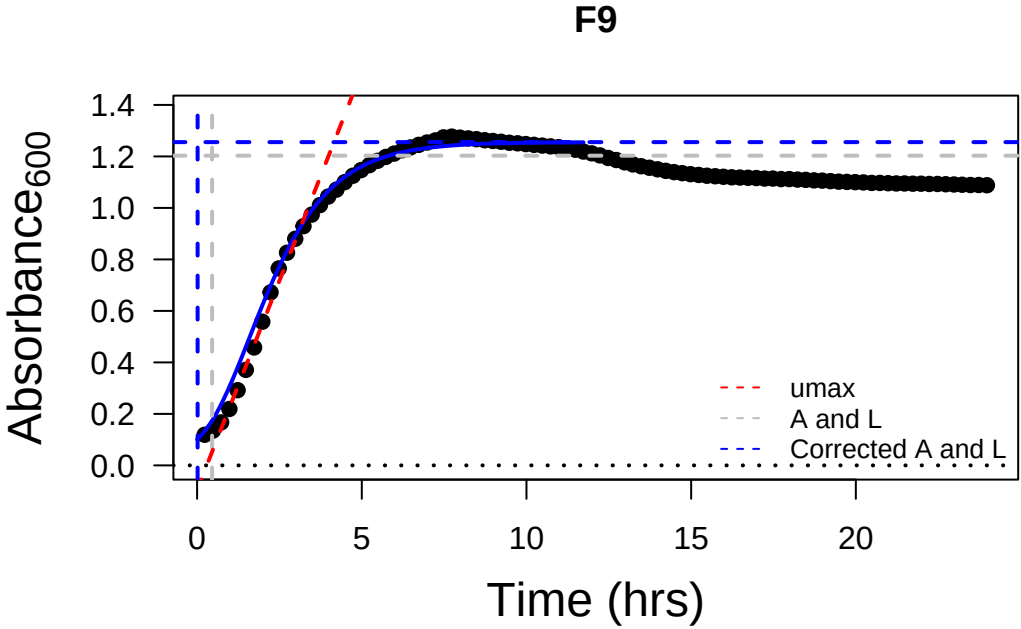

F4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

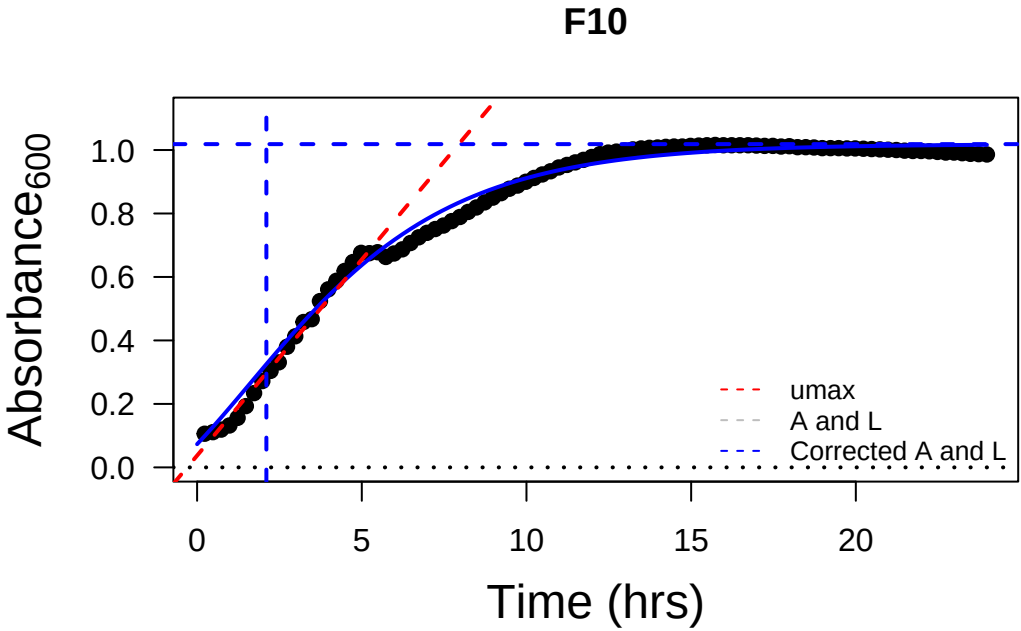

F5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

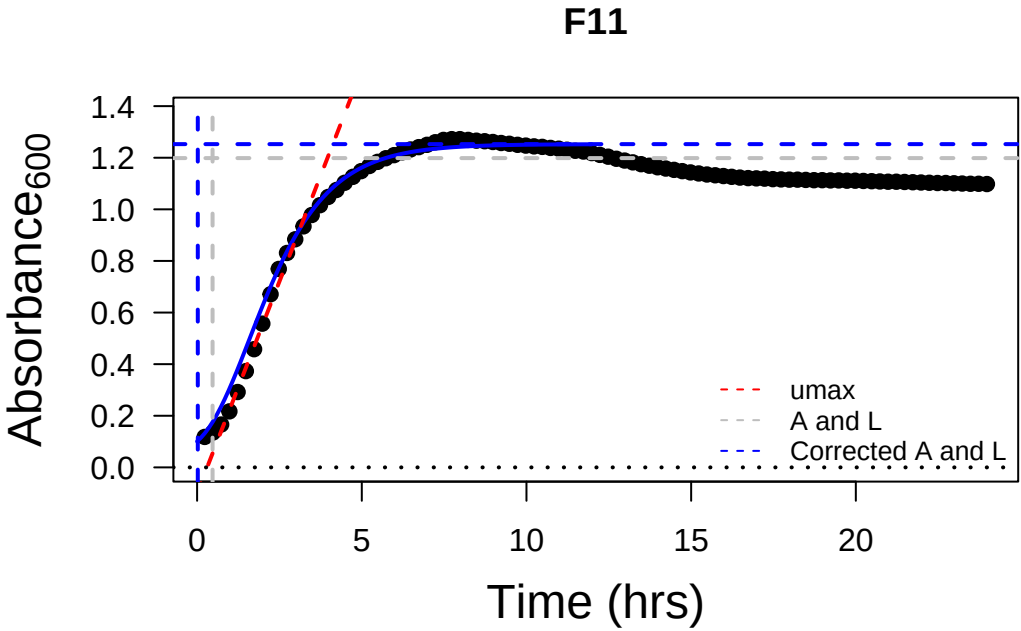

F6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

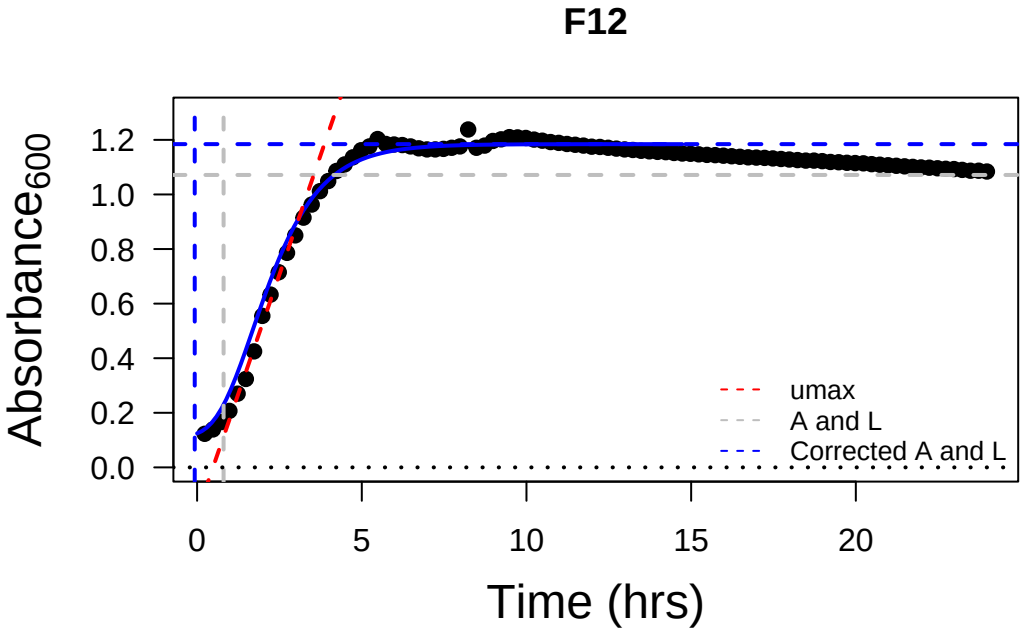

F7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

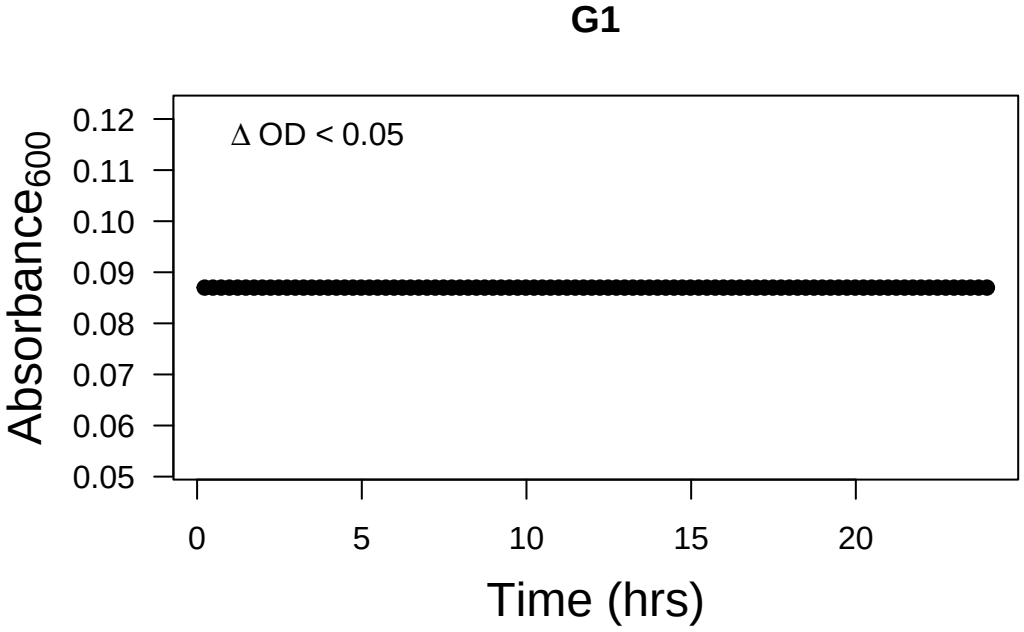

F8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

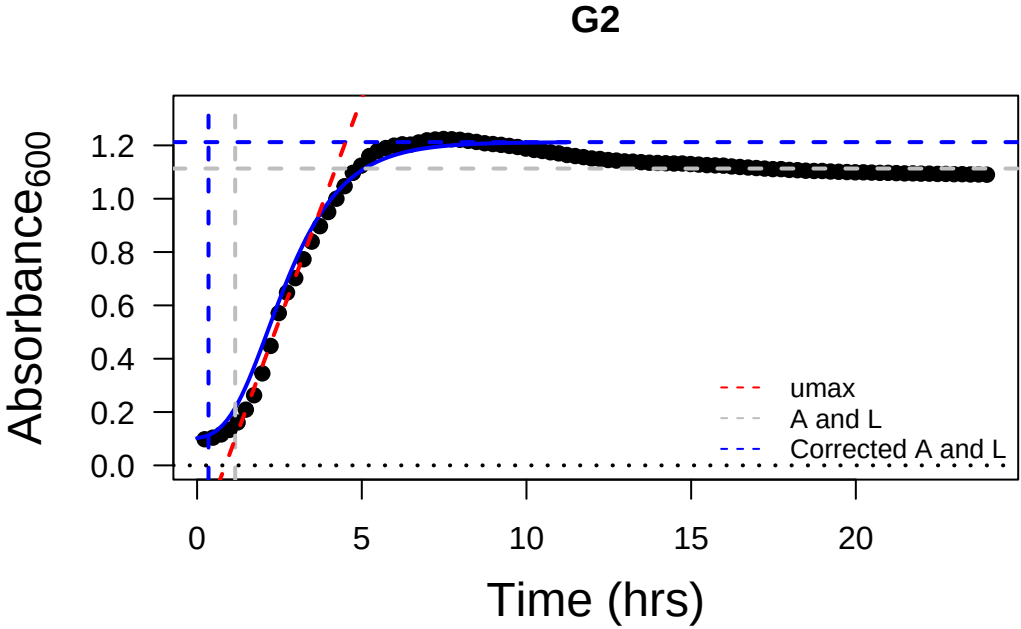

F9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

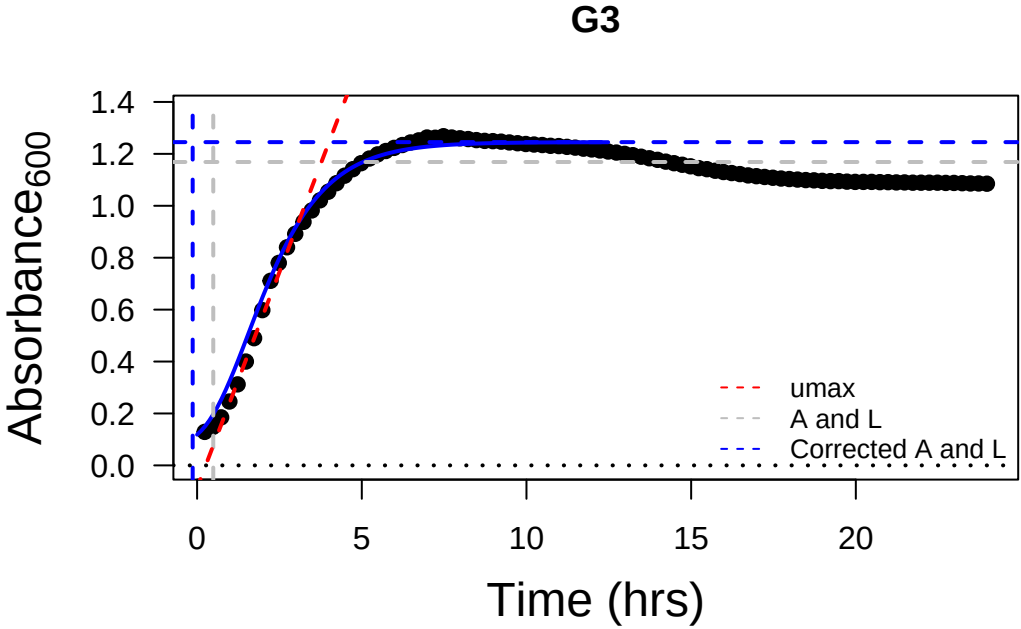

F10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

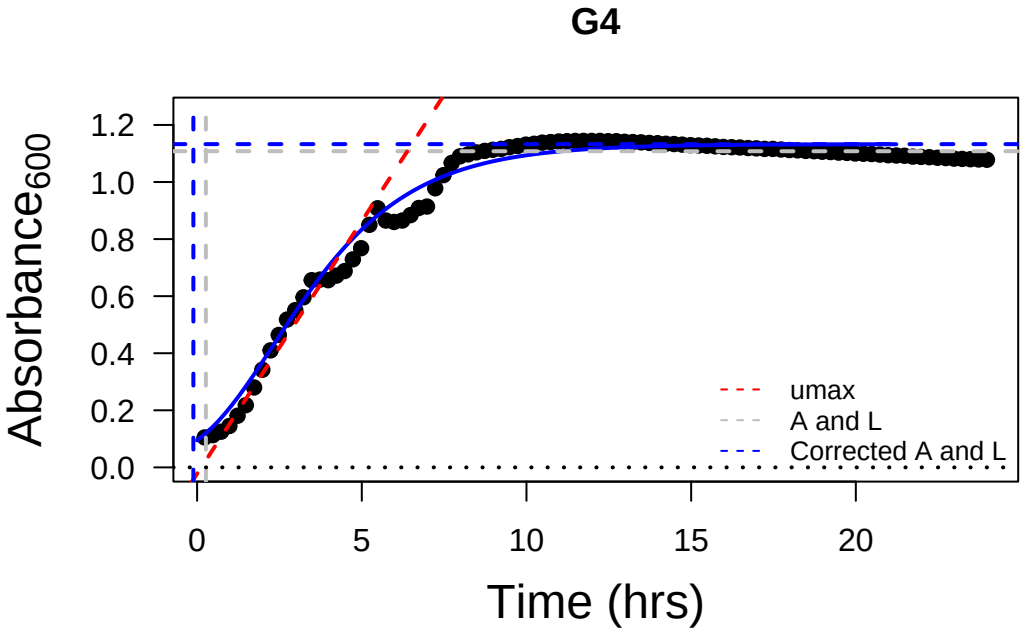

F11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

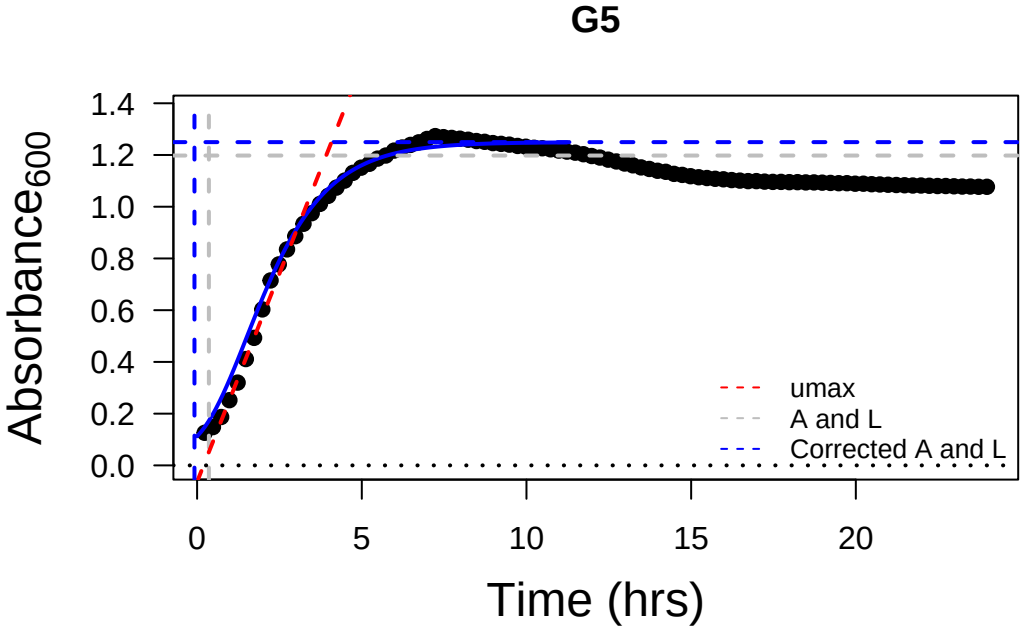

F12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

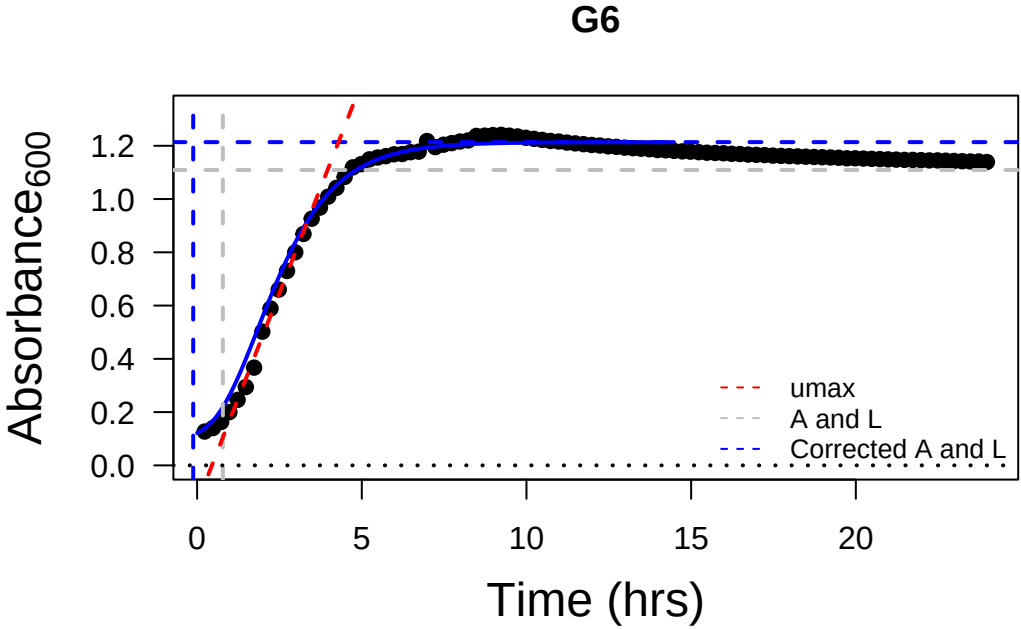

G1
Δ OD < 0.05
Absorbance600
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0
5
10
15
20
Time (hrs)

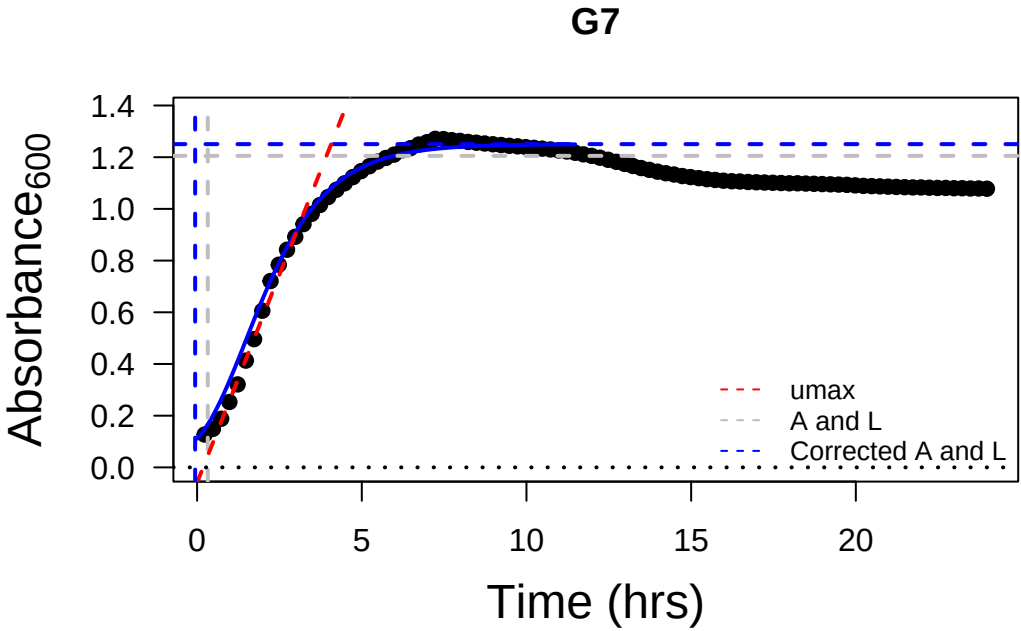

G2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

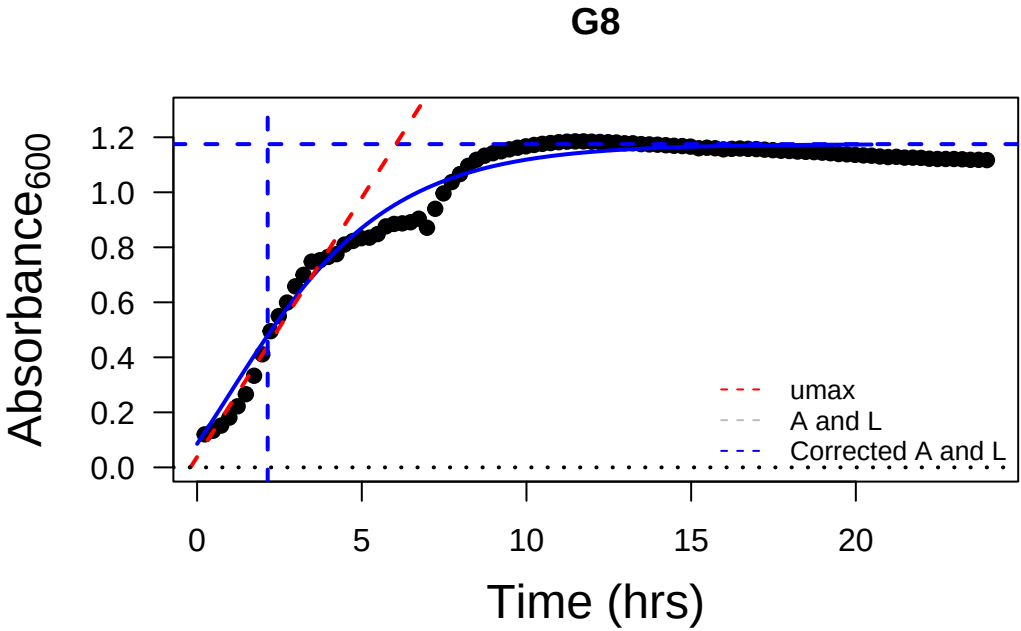

G3
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

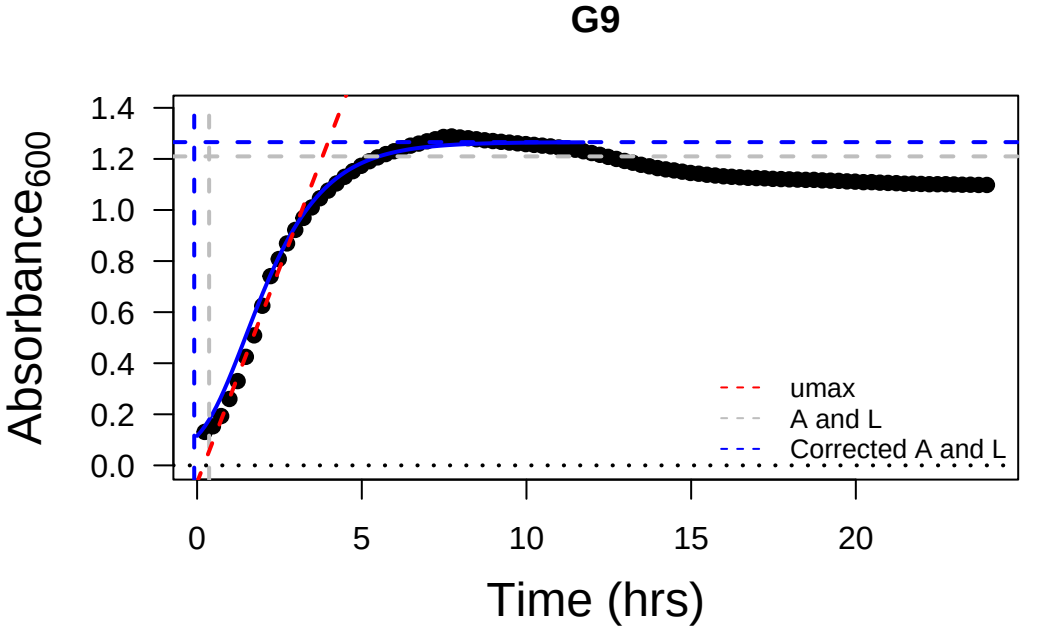

G4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

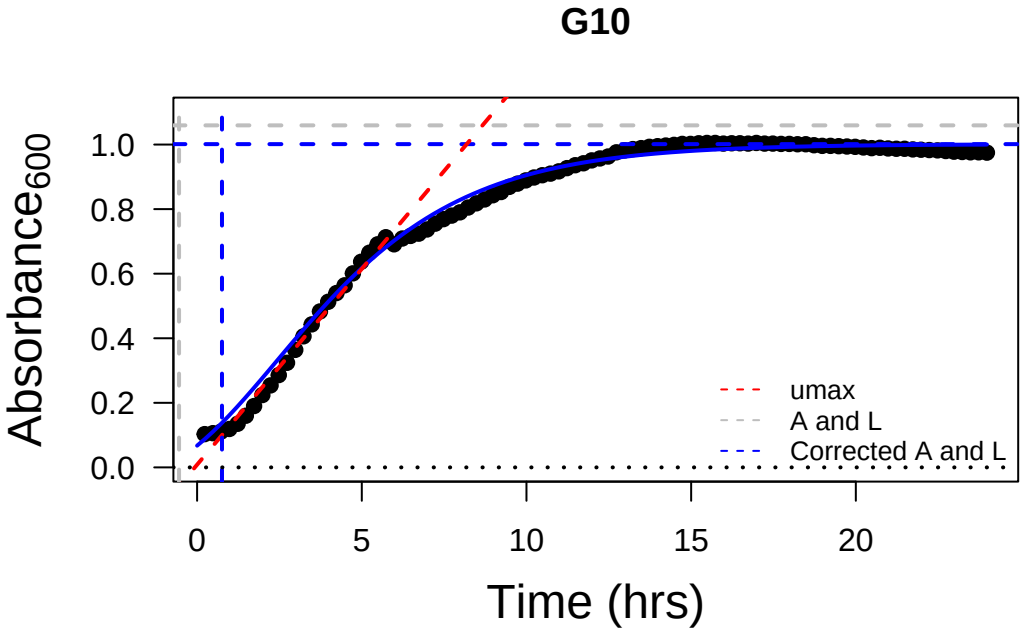

G5
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

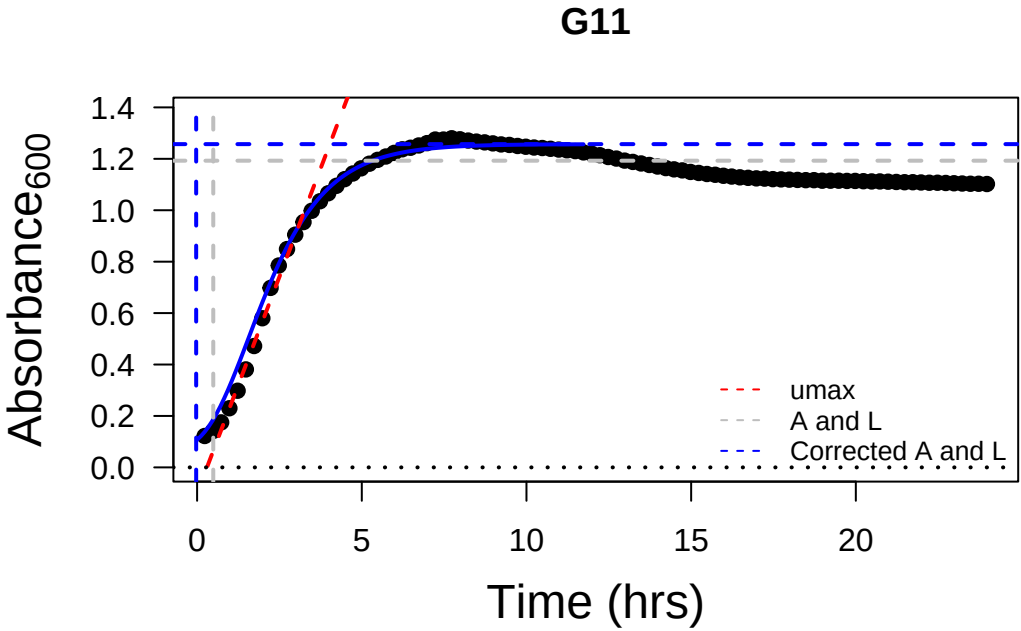

G6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

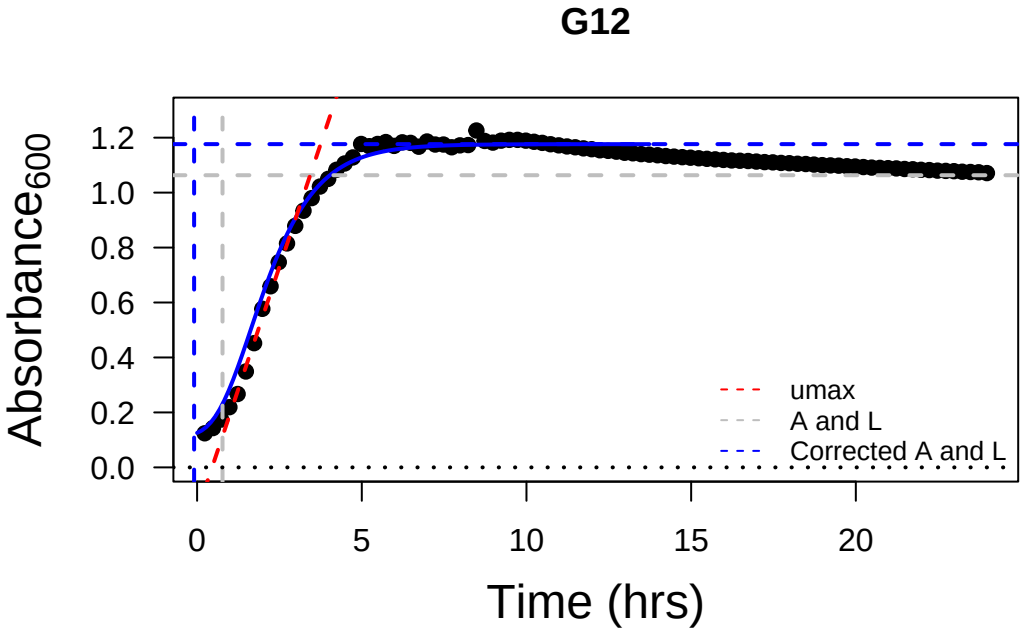

G7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

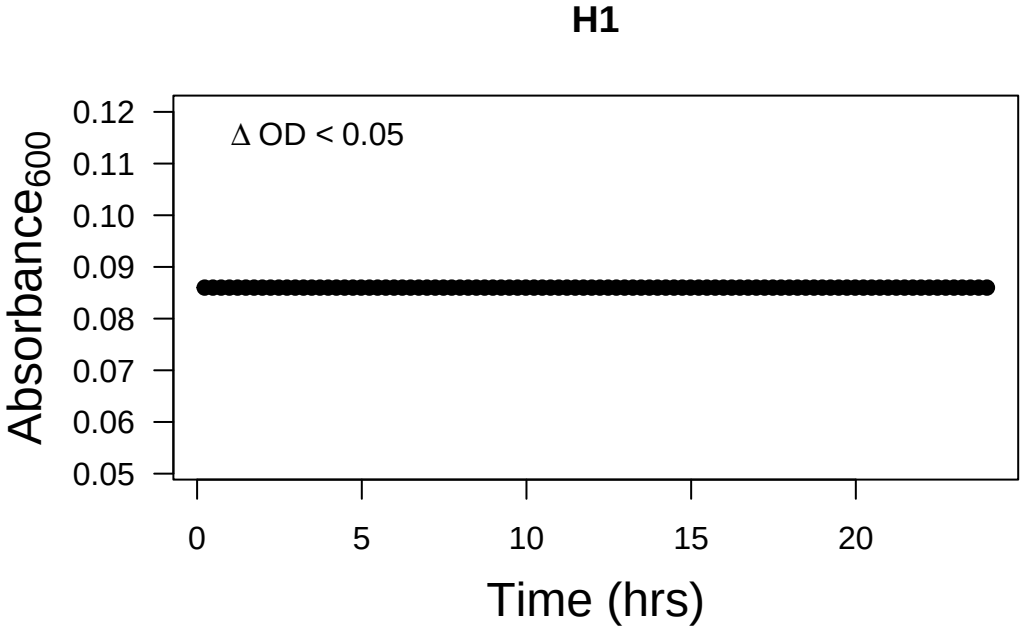

G8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

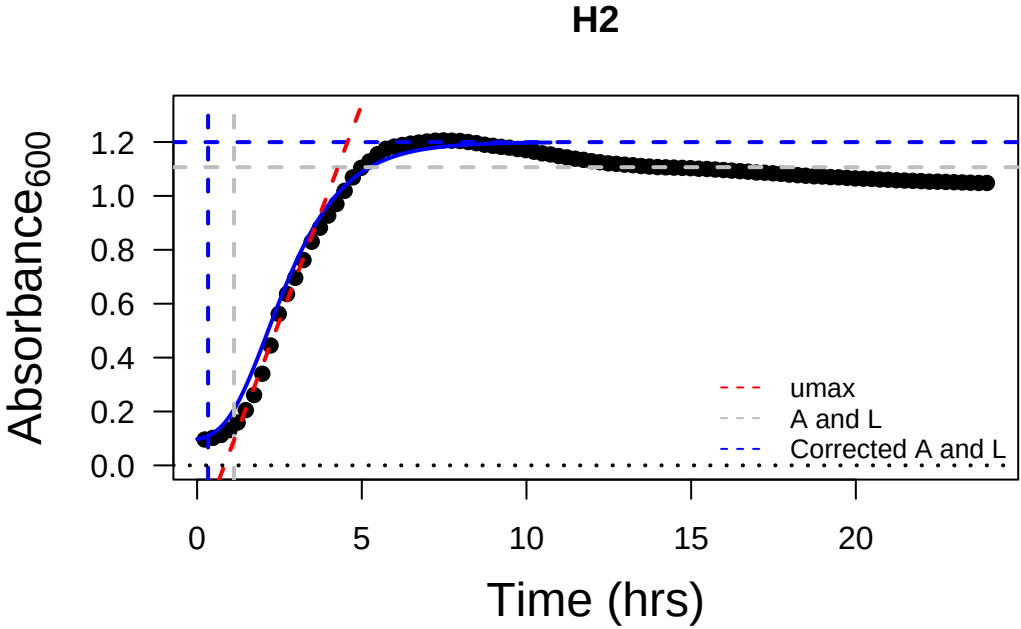

G9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

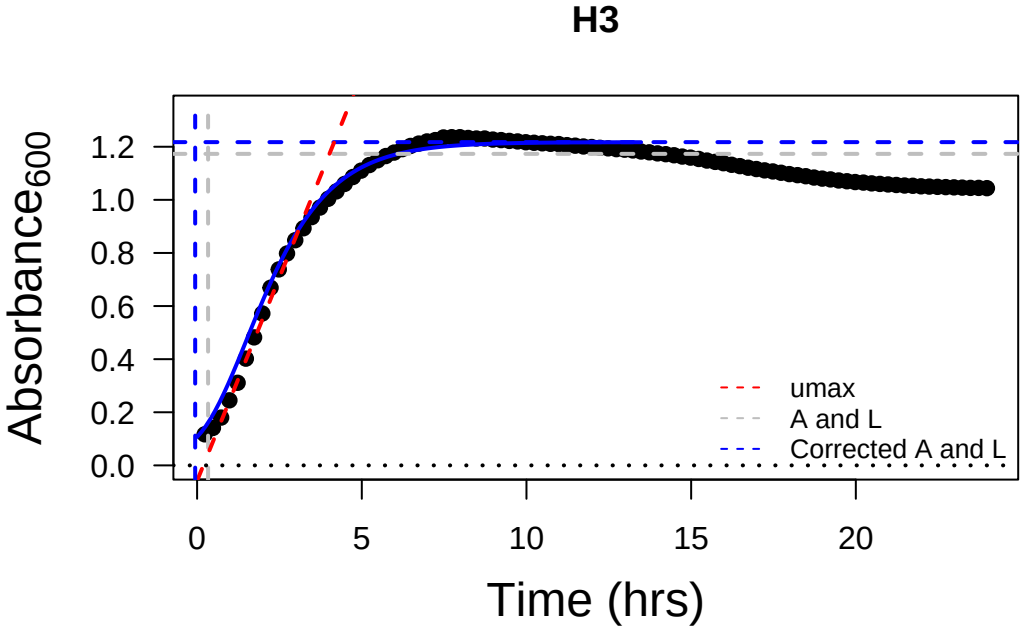

G10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

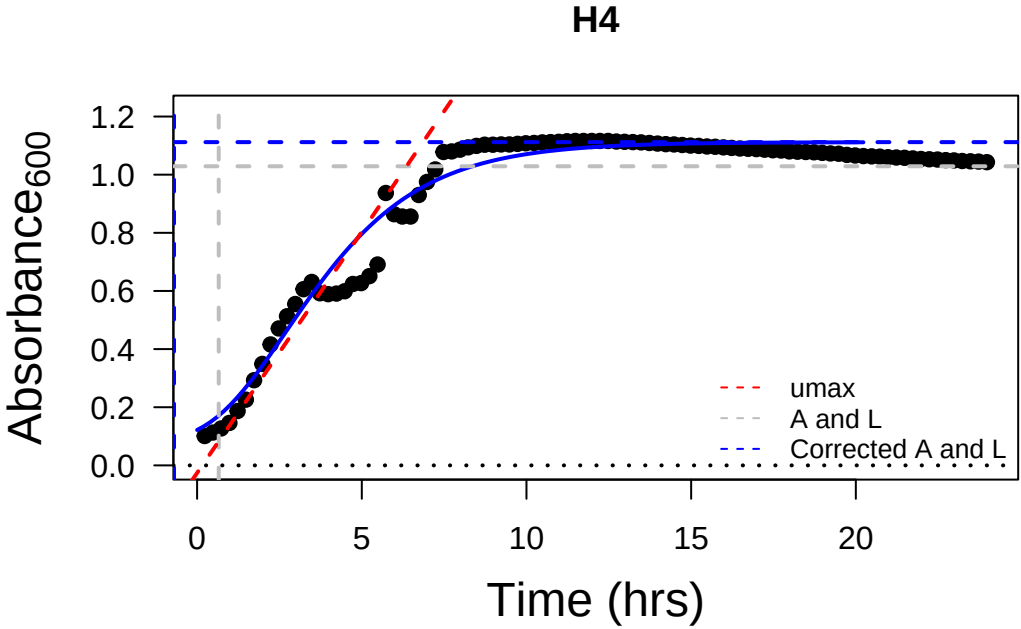

G11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

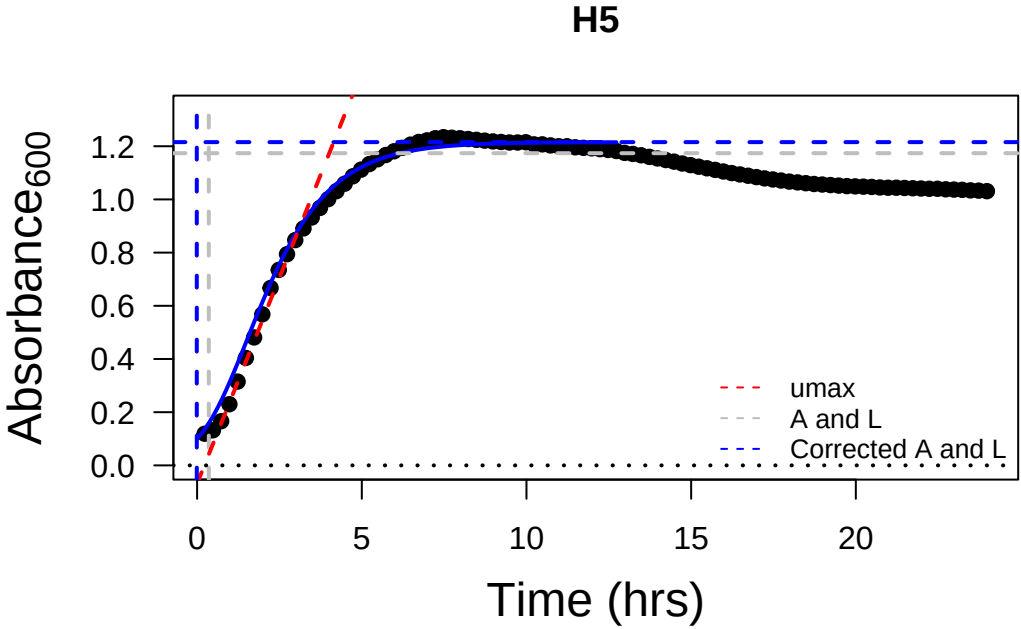

G12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

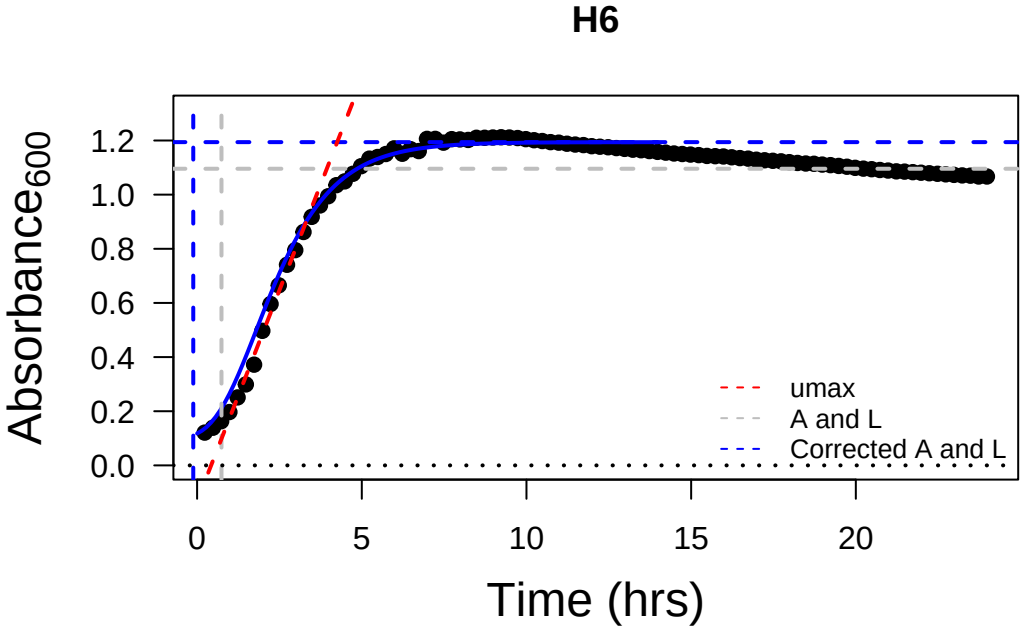

H1
Δ OD < 0.05
Absorbance600
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0
5
10
15
20
Time (hrs)

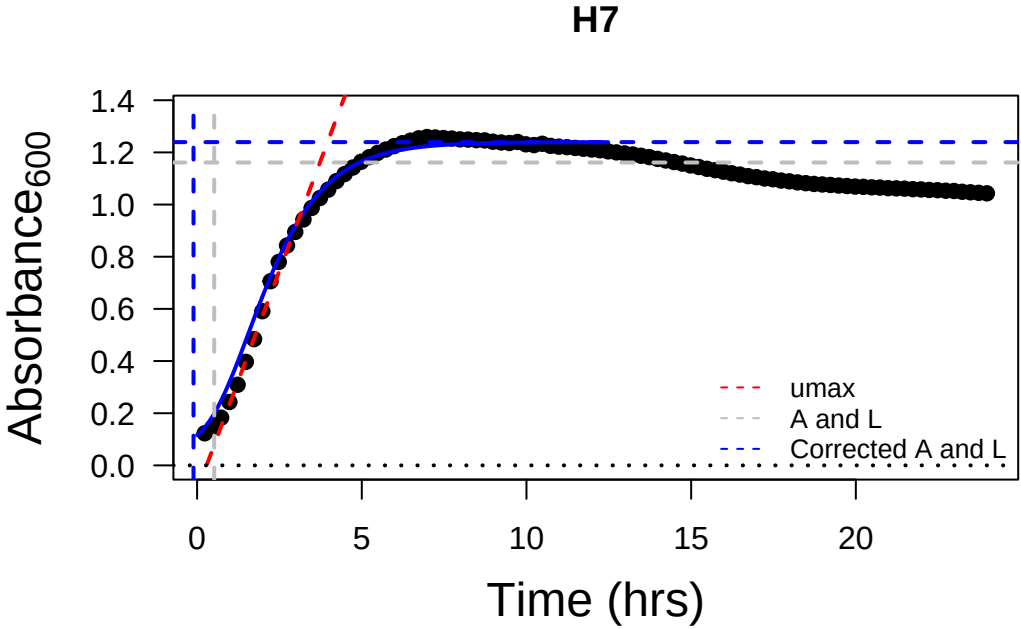

H2
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

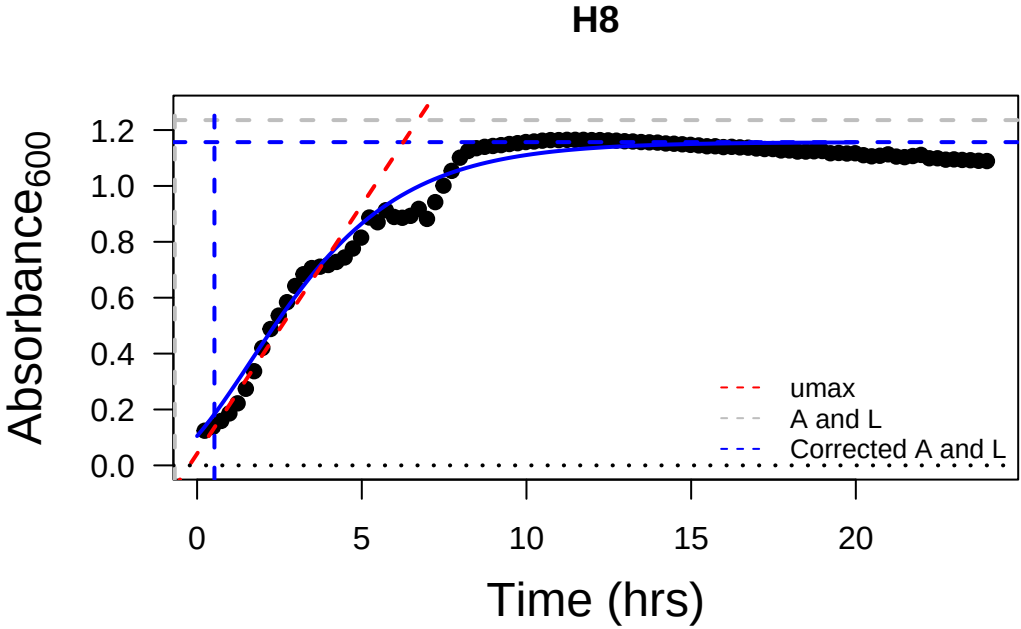

H3
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

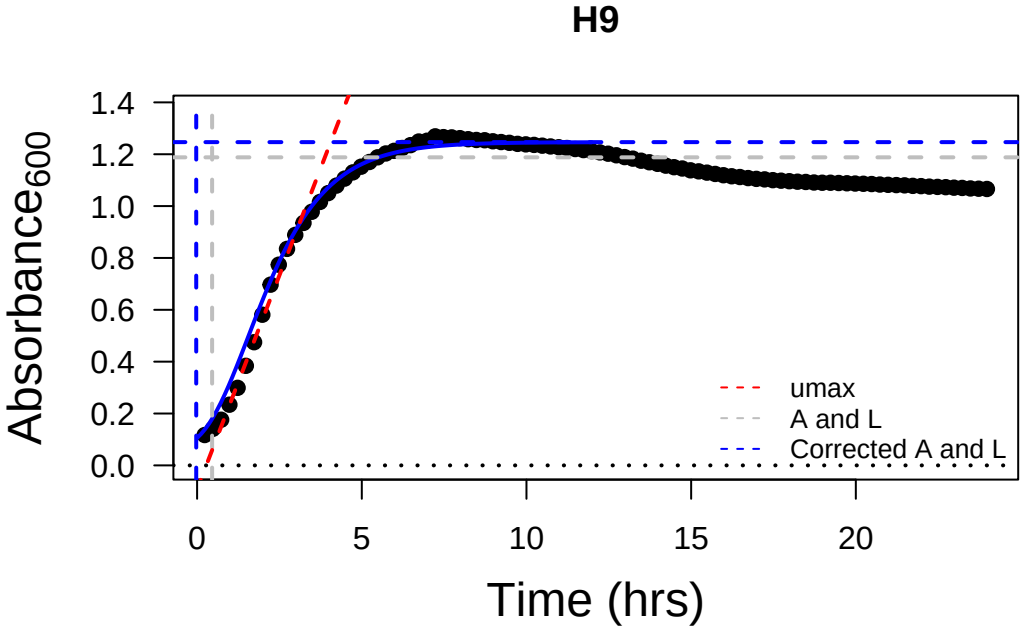

H4
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

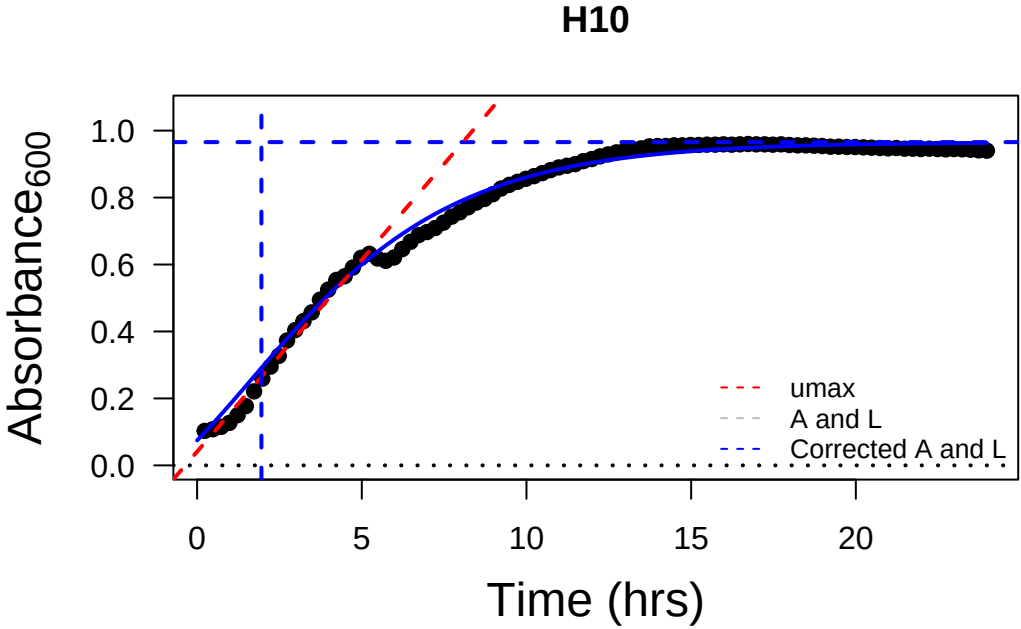

H5
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

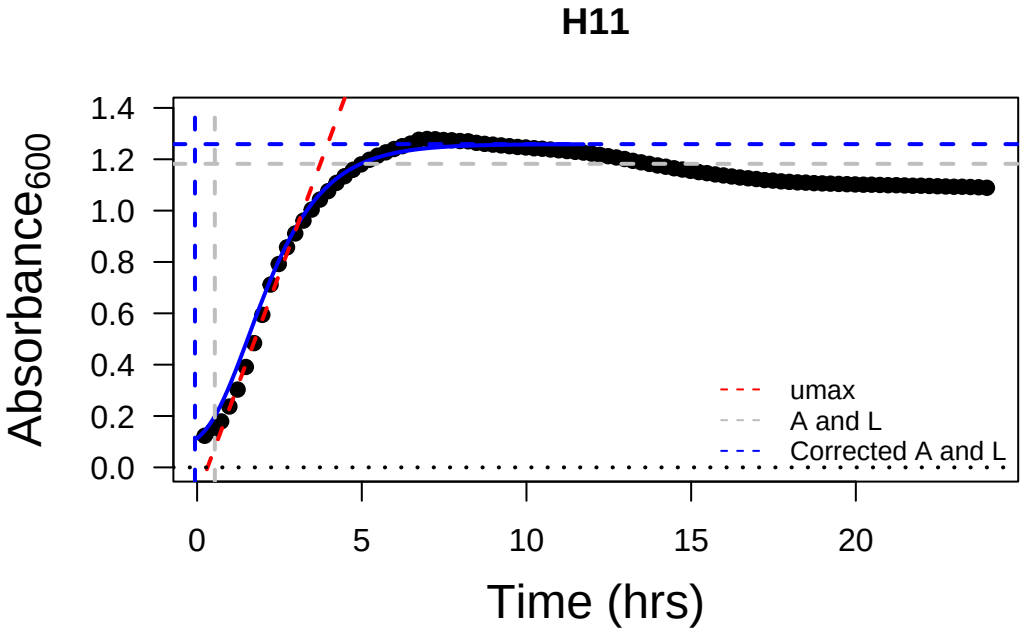

H6
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L

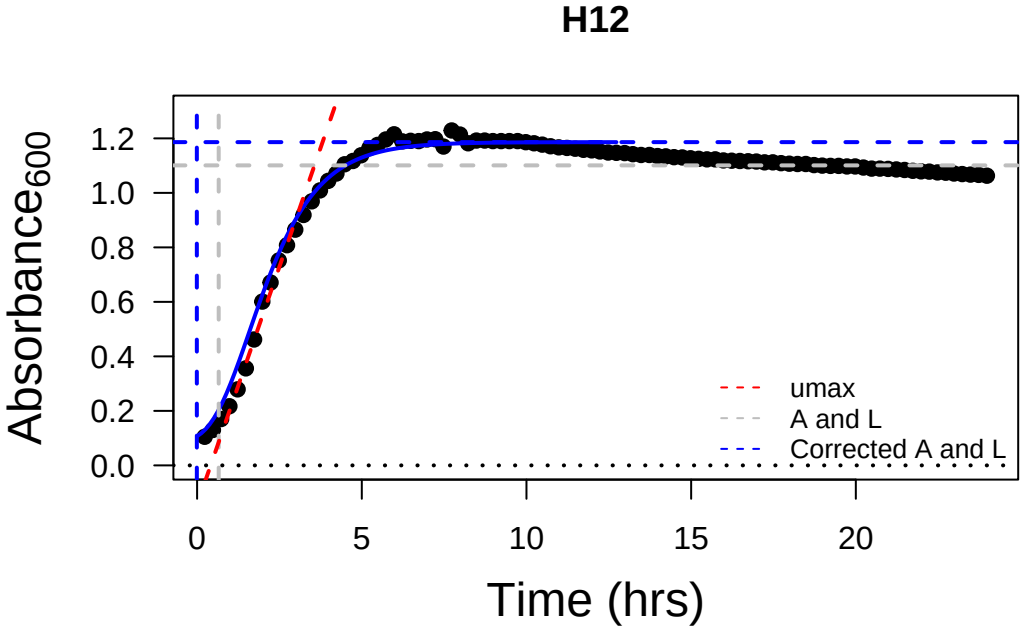

H7
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L


H8
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L


H9
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L


H10
Absorbance600
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L


H11
Absorbance600
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L


H12
Absorbance600
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
Time (hrs)
umax
A and L
Corrected A and L